



Systematic comparison of software agents and Digital Twins: differences, similarities, and synergies in industrial production

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Abstract

To achieve a highly agile and flexible production, a transformational shift is envisioned whereby industrial production systems evolve to be more decentralized, interconnected, and intelligent. Within this vision, production assets collaborate with each other, exhibiting a high degree of autonomy. Furthermore, information about individual production assets is accessible throughout their entire life-cycles. To realize this vision, the use of advanced information technology is required. Two commonly applied software paradigms in this context are Software Agents (referred to as Agents) and Digital Twins (DTs). This work presents a systematic comparison of Agents and DTs in industrial applications. The goal of the study is to determine the differences, similarities, and potential synergies between the two paradigms. The comparison is based on the purposes for which Agents and DTs are applied, the properties and capabilities exhibited by these software paradigms, and how they can be allocated within the Reference Architecture Model Industry 4.0. The comparison reveals that Agents are commonly employed in the collaborative planning and execution of production processes, while DTs are generally more applied to monitor production resources and process information. Although these observations imply characteristic sets of capabilities and properties for both Agents and DTs, a clear and definitive distinction between the two paradigms cannot be made. Instead, the analysis indicates that production assets utilizing a combination of Agents and DTs would demonstrate high degrees of intelligence, autonomy, sociability, and fidelity. To achieve this, further standardization is required, particularly in the field of DTs.

Keywords Software Agent · Digital Twin · Industry 4.0 · Production · Systematic comparison · PRISMA · RAMI 4.0

Abbreviations

AAS	Asset administration shell
CD	Complexity degree
DT	Digital Twin
MAS	Multi-agent system
PRISMA	Preferred reporting items for systematic reviews and meta-analyses
RAMI 4.0	Reference architecture model industry 4.0

Introduction

Industrial enterprises face demanding challenges concerning the competitiveness and sustainability of their operations (Aheleroff et al., 2022). Global competition, resource scarcity, and increasing customer demands for individualized products necessitate shorter product development cycles and more agile production environments (Ashtari Talkhestani et al., 2019). To tackle these challenges, it is envisioned that production systems become increasingly interconnected, cooperative, and autonomous. Assets such as production machines and products will exhibit a high degree of intelligence, supported by enhanced data collection and information processing capabilities (Monostori, 2014). This vision is commonly referred to as Industry 4.0 (Sakurada & Leitão, 2020). The realization of this vision is expected to yield several benefits, including the optimization of production

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processes, increased product individualization, improved robustness of production systems, enhanced system transparency, and increased resource efficiency (Cardin, 2019; Monostori, 2014). To realize these benefits and implement autonomous, cooperative, and interconnected production systems, suitable use of information and communication technology is required.

The Digital Twin (DT) and its standardized implementation, the Asset Administration Shell (AAS), are often seen as one of the most prominent paradigms to enable the process of digitization within Industry 4.0 (Sakurada et al., 2022a). Software Agents, particularly Multi-Agent Systems (MAS), are often viewed as a key enabler for autonomous and self acting DTs (Sakurada et al., 2022a; Vogel-Heuser et al., 2020; Sakurada & Leitão, 2020) as well as an enabler for realizing the full potential of Industry 4.0 (Karnouskos et al., 2019).

Therefore, this work investigates two software paradigms commonly viewed as key enablers of Industry 4.0: Software Agents (referred to as Agents) and Digital Twins (Karnouskos et al., 2020).

Agent paradigm

An Agent is “[...] a computer system, situated in some environment, that is capable of flexible autonomous action in order to meet its design objectives” (Jennings et al., 1998). Agents are commonly associated with properties and capabilities such as autonomy, reactivity, proactivity, deliberativeness, persistence, encapsulation, and communicative abilities (VDI, 2010), which are required for interacting with one another in so called MAS. There are a number of advantages that distinguish Agent-based automation solutions from conventional, centralized concepts (Jennings & Bussmann, 2003; VDI, 2010):

- Conceptual advantages arise from inherent problem decomposition during engineering
- No single point of failure or communication bottleneck
- Interaction with the environment and utilization of local knowledge, reducing communication traffic.
- Flexible communication paths and organizational relationships
- The distribution of computing power enables high scalability, facilitating flexible and efficient adaptation to changing production requirements and workloads.
- Adaptive and flexible behavior, allowing for dynamic reconfiguration and coordination of Agents in response to changing conditions and requirements in the industrial environment
- Independent operation of Agents, enabling MAS to compensate failures of other Agents
- By coordinating efforts among Agents, decentralized optimization of production processes can be achieved,

improving resource allocation, task allocation, and scheduling

Digital twin paradigm

The concept of the DT as a virtual representation of an asset has first been described by Grieves (2014). Over the past decade, various definitions have surfaced in an attempt to delineate and encapsulate this concept more precisely. NASA defined the DT “as an integrated multiphysics, multiscale, probabilistic simulation of an as-built vehicle or system that uses the best available physical models, sensor updates, fleet history, etc., to mirror the life of its corresponding flying twin” (Glaessgen & Stargel, 2012). Moreover, Madni et al. (2019) distinguish the DT from a Computer Aided Design/Computer Aided Engineering model. In this context, they emphasize, among other things, that the DT is a “...specific instance...” of a physical twin that “...combines data...” and is used to “...observe system performance...” and “...enable traceability...” across different life cycle phases. In this work, the definitions introduced by Tao et al. (2018) and Kritzinger et al. (2018) are referred to and used as a guideline in order to establish a common understanding of this concept. Both Tao et al. (2019) and Kritzinger et al. (2018) have presented two definitions and descriptions of the DT, its components, and the objectives of its usage, which are frequently cited and acknowledged by the academic community. Accordingly, the aim here is to cultivate a shared understanding of the DT concept through such referencing. Tao et al. (2018) define the DT as a digital representation of an asset throughout its life cycle, capable of reflecting its static properties, its dynamic behavior as well as its current condition. In this context, a DT can mirror the real world object or system in real time, thus being beneficial for use cases like monitoring, forecasting, analysis and optimization purposes as well as control (Kritzinger et al., 2018; Tao et al., 2018). Interaction between the digital representation, consisting of relevant digital models, and the physical asset is achieved by establishing a bidirectional and automated data flow (Kritzinger et al., 2018). This enables continuous model improvement and synchronization with the real world asset or system (Tao et al., 2018). The DT includes static data such as technical documentation, as well as dynamic data such as mathematical or simulation models that reflect the asset’s behavior (Kritzinger et al., 2018).

In this work the DT is considered to take both, a passive as well as an active role. In the passive role, similar to the digital shadow concept proposed by Kritzinger et al. (2018), synchronization of digital models with the physical asset occurs by establishing a unidirectional data flow. During this process, data is collected, stored, and used for condition monitoring, analysis, or optimization using the existing digital models. However, no immediate or simultaneous influence is exerted on the physical asset. In contrast, in the active role,

the DT is fully integrated by establishing a bidirectional data flow and has control over the physical asset. Both roles are taken into account by the authors of this paper for the systematic comparison of the paradigms.

The advantages of using DTs are the following (Kritzinger et al., 2018; Tao et al., 2018, 2019; Pires et al., 2021; Mashaly, 2021):

- Efficient design and development of Industry 4.0 conform production systems by means of virtual modeling and simulation of products, processes, and systems in the early stages of development or during reconfiguration.
- Facilitating cooperation among designers, engineers, operators, and manufacturers during the engineering phase, thanks to a digital system model.
- Improving operational efficiency by providing real time information about the condition, performance, and behavior of physical assets.
- Enabling condition monitoring using real time operational data and historical data
- Supporting operational decisions by visualizing operational data.

Motivation

While the previous conceptual descriptions of Agents and DTs suggest certain differences between the two paradigms, a growing number of applications of Agents and DTs for similar purposes can be observed in the literature. For instance, both Agents and DTs have been employed in production scheduling (Klein et al., 2018; Tliba et al., 2023), fault diagnosis (Xu et al., 2019; Seng Ng & Srinivasan, 2010), reconfiguration of manufacturing processes (Leng et al., 2020; Rodrigues et al., 2018), and the control of production resources at the control device level (Leng et al., 2020; Schutz et al., 2011). Furthermore, similar sets of requirements for realizing the vision of Industry 4.0 are addressed by approaches utilizing both paradigms (Seitz et al., 2021; Redelinghuys et al., 2020). Such similarities in the purposes served by Agents and DTs can be indicative of general similarities between the two paradigms. A systematic comparison of Agents and DTs allows for a more in-depth investigation of the extent of this similarity. By identifying applications of the two paradigms that exhibit similarities, researchers specializing in Agents, for example, can gain insights into potential solutions and best practices for a given problem from the realm of DTs, and vice versa. This approach can help researchers to avoid pitfalls identified in one paradigm when working with the other. Analogously to similarities between Agents and DTs, significant differences between the two paradigms can be identified through systematic comparison, providing guidance to researchers, especially those

with limited experience, on which paradigm is best suited for solving a particular problem.

In addition to the applications of Agents and DTs for similar purposes discussed previously, some publications employ Agents and DTs synergistically. For instance, Vogel-Heuser et al. (2021) utilize DTs as a standardized knowledge base for Agents, while Ashtari Talkhestani et al. (2019) present an architecture for an intelligent DT and employs Agents to facilitate co-simulation between different DTs. Another example of the concurrent and synergistic use of Agents and DTs involves the application of design patterns from the Agent paradigm to develop “proactive” AASs, resulting in DTs exhibiting Agent-like capabilities, such as proactivity (López et al., 2023). A systematic comparison of Agents and DTs will also aid in establishing guidelines on which applications can benefit from a synergistic utilization of Agents and DTs, where the strengths of one paradigm compensate for the weaknesses of the other. Analogously to synergies, dissonances between the two paradigms can be identified, resulting in guidelines on how not to apply Agents and Digital Twins simultaneously. Based on a systematic comparison, a starting point can be set for a strategic co-evolution or even a prospective integration of the two paradigms.

To summarize the previous paragraphs, three observations should be highlighted:

1. Agents and DTs are sometimes applied for similar purposes.
2. There are use cases in which Agents and DTs are used synergistically.
3. The formal definitions of Agents and DTs indicate significant differences between the paradigms.

This work further explores these observations to determine the similarities and differences of Agents and DTs within actual implementations in industrial automated production systems. By focusing on actual implementations of Agents and DTs, purely conceptual reports are disregarded in this work. This is done to obtain insights into the capabilities required for industrial applications and evaluate their feasibility. The exclusion of purely theoretical reports is based on the premise that concepts can be proposed regardless of their applicability in or suitability for industrial practice.

Therefore, the research questions pursued in this work can be summarized as follows:

1. What are the capabilities and properties exhibited by Agents and DTs?
2. What purposes do Agents and DTs serve within industrial production systems?
3. What are differences, similarities, and potential synergies between Agents and DTs in industrial practice?

To address these research questions, a systematic analysis of reports describing applications of Agents and DTs is conducted. Based on the analysis, the properties and capabilities exhibited by Agents and DTs are identified. Additionally, each application of Agents and DTs is allocated to the elements of the *Reference Architecture Model Industry 4.0* (RAMI 4.0), i.e., the layers, hierarchy levels and life cycle phases, which the respective report primarily addresses. Finally, the purposes fulfilled by the Agents and DTs are determined for each report. The precise method for conducting these analyses is described in “[Methodology](#)” section.

The insights obtained from the conducted analyses can serve as a basis for practitioners, looking for the most suitable software paradigm to solve specific problems. The focus on actual implementations thereby allows to move beyond conceptual comparisons and investigate what is actually implemented in practice. This in turn can serve as a base for discussions on the conceptual definitions of the paradigms’ definitions. In fact, it has been suggested by other authors that “*the Digital Twin would benefit from a more detailed comparison and review in the context of similar and connected fields*” (Jones et al., 2020) and that “*some authors use the term [Digital Twin] merely as a catchphrase*” (Sjarov et al., 2020), indicating the lack of a clear and common definition and standardization, which is investigated within this work. Finally, the results of this analysis provide insights into potential synergies between the two paradigms, thereby identifying future research opportunities.

Related works

Several reviews have been published aiming to describe characteristics of Industry 4.0-related paradigms and/or a comparison thereof. De Paula Ferreira et al. (2020) review simulation in Industry 4.0 and derive “design principles” to classify simulation paradigms, including Agents and DTs. However, they do not conduct a detailed comparison between Agents and DTs (de Paula Ferreira et al., 2020). Müller et al. (2021) present a list of abilities and characteristics to describe industrial autonomous systems, which is then used to exemplify a single use case (Müller et al., 2021). Jones et al. (2020) aim at characterizing the DT paradigm in depth, providing a description of its characteristics, but do not apply the list to compare different DT applications (Jones et al., 2020). Cohen et al. (2023) conduct a review of artificial intelligence applications in DTs and offer recommendations on which artificial intelligence techniques are suitable for different purposes of DT applications. Agents are not within the scope of Cohen et al. (2023). Lehmann et al. (2023) explore the symbiotic relationship between intelligent DTs and MAS. Lehmann et al. (2023) highlight the similarities in terms of specific characteristics of MAS, such as being active, online, goal seeking,

and anticipatory. However, their comparison between DTs and MAS is qualitative, and no systematic literature review has been conducted.

Furthermore, research has been devoted to analyzing the alignment of Agents and DTs with RAMI 4.0. Verbeet and Baumgärtel (2020) propose an approach to implement an autonomous Industry 4.0 system using Agents and classify the mandatory and optional capabilities of Agents base on the “layers” and “hierarchy levels” axes of RAMI 4.0. Melo et al. (2023) align the DT concept with the different dimensions of RAMI 4.0 and consider the capabilities of MAS to support the development of Digital Twin ecosystems.

As mentioned earlier, numerous literature reviews exist, but none conduct a systematic comparison of Agents and DTs to identify differences, similarities, and synergies as well as common purposes.

Contributions of this systematic comparison

The analysis of related works reveals that the research questions stated in “[Motivation](#)” section have not been addressed in previous studies. Therefore, this systematic comparison of Agents and DTs offers the following contributions:

1. Systematic literature review following the most recent Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement to identify relevant reports concerning the application of Agents and DTs in industrial automated production systems.
2. Comparison of Agents and DTs based on their capabilities and properties as well as their allocation within RAMI 4.0
3. Analysis of use cases of Agents and DTs including the determination of the purposes for which they are applied.
4. Identification of differences, similarities, and synergies between Agents and DTs.

The remainder of this work is structured as follows: “[Methodology](#)” section describes the methodology employed to (1) identify relevant reports regarding industrial applications of Agents and DTs and (2) systematically compare them. “[Results](#)” section presents the results of the systematic literature review, along with the determination of system properties, capabilities, and purposes. In “[Discussion](#)” section, a discussion is conducted on the applied methods and the presented results.

Methodology

The methodology applied in this work consists of two main parts: a systematic review process to identify relevant reports from the existing literature (“[Methodology](#)” section) and a method to facilitate a systematic comparison

Table 1 Search query used to identify relevant records

Field	Title, Abstract, Keywords
Search term	(agent* OR mas OR digital-twin OR “digital twin” OR twin OR “administration shell” OR aas) AND (industr* OR produc* OR manufact* OR process) AND automa* AND (“case study” OR application OR implement*)
Excluded	blockchain OR health* OR human OR bio* OR city OR home OR vehicle

of the reports identified within the systematic review process (“[Methodology for the systematic determination of capabilities, properties, and purposes](#)” section and subsequent sections).

Method of the systematic literature review

The systematic literature review adheres to the PRISMA 2020 statement (Page et al., 2021). The following sections outline the search query, the reviewing process, and the criteria for excluding or including records. In this context, the term “record” refers to publications obtained from the search engine based on the search query, while “report” refers to those records that met the inclusion criteria and were subsequently analyzed concerning the application of Agents or DTs.

To identify potentially eligible records, a systematic search was conducted on June 15, 2023, using the Scopus © database (Elsevier, 2022). The search query is presented in Table 1.

Results were limited to English and German publications. Since the Industry 4.0 framework was introduced at the Hanover fair in 2011 (Geisberger et al., 2011), records dating back to 2011 were included. However, this literature review focuses on recent works in the field, i.e., published in the last 5 years, to provide an up-to-date analysis. This is also due to the fact that a significant rise in publications in the field can be seen starting from 2017. To achieve this up-to-date result, the latest works from 2023 are included without limitations on citations since these works have not had time to accumulate such citations. With an increase in the years since publication, an increasing threshold on the number of required citations was applied to ensure that only high quality research is included. Further, to achieve the focus on recent research, a significant increase, i.e., from 5 to 10, in the applied threshold for required citations of works that are older than 5 years has been applied (see Table 2). However, instead of excluding such works all together, the latter acknowledges that the survey should include seminal works from the period of 2011 to 2016 that had a significant influence on the field.

Table 2 Minimum number of citations by year of publication

Year of publication	Minimum number of citations
2022, 2023	0
2021	1
2020	2
2019	3
2018	4
2017	5
older than 2017	10

The reviewing process was structured into three phases, with the same inclusion and exclusion criteria applied throughout the entire process. For a record to be included, it had to fulfill all of the following criteria; otherwise, it was excluded. These three inclusion criteria were evaluated simultaneously:

1. The record must primarily focus on either Agents or DTs.
2. The area of application must be industrial automated production systems.
3. The record must include an application, implementation, or case study. Purely conceptual or review papers were excluded. The implementation described should provide sufficient detail to allow characterization using the method presented in “[Methodology for the systematic determination of capabilities, properties, and purposes](#)” section.

The records underwent analysis in three consecutive phases. In phase 1, only the title of each record was analyzed to eliminate those that were clearly out of scope. During phase 2, the title, abstract, and keywords of the remaining records were screened. Possible ratings included 0 (clearly out of scope), 1 (unclear), and 2 (criteria for inclusion fulfilled). If a record received a rating of 1, it was independently screened again by a different reviewer. For all records rated 1 or 2 in phase 2, the full text was obtained and analyzed in phase 3. The analysis followed the method outlined in “[Methodology for the systematic determination of capabilities, properties, and purposes](#)” section. If the analysis of the full text revealed that any of the inclusion criteria were not

Table 3 Definition of purpose, properties, and capabilities

Aspect	Definition
System purpose	The system purpose describes the future state the system should achieve. (Weilkiens, 2014)
System properties	System properties characterize the system in its target state. They represent and group the system's capabilities on an abstract level. (Froese & Kleinblotekamp, 2019)
System capabilities	System capabilities extend a system's properties and describe the power or skill of the system to fulfill the purpose of the system (Fay et al., 2019). It encompasses the expression ability as well

met, the respective record was excluded during the full text screening in phase 3.

To prevent bias during the first two phases, information regarding authors, journals/conferences, and citation counts was omitted. Additionally, different reviewers were assigned to screen each record in every phase. To ensure consistent decisions regarding inclusion and exclusion, a joint analysis of at least 15 reports was conducted in each phase. Furthermore, 10% of the included reports were independently screened by two reviewers in phase 3.

Methodology for the systematic determination of capabilities, properties, and purposes

To compare Agents and DTs, the purposes, properties, and capabilities of both paradigms were used as comparative aspects. Table 3 presents the definitions of the terms purpose, property, capability as interpreted in this work. The terms have been adapted from model-based systems engineering (Weilkiens, 2014) serving to characterize complex systems. In this context, the term property is utilized to characterize the system in its target state, offering an overview or abstract representation of the system's potential in achieving its intended objectives. On the other hand, capability, synonymous with functionality, delineates the specific skills or abilities that the system must possess, and thus, is more granular and precise compared to properties.

The sets of capabilities, properties, and purposes which were attributed to the applications of Agents and DTs were derived from a literature analysis and iteratively adjusted in the initial screening. In total 26 capabilities were identified as well as four properties and nine categories of purposes. The process of identifying the capabilities, properties, and purpose of the Agents and DTs in a given report is described in “[Identification of capabilities](#)”, “[Identification of properties](#)”, and “[Identification of purposes](#)” sections, respectively.

Identification of capabilities

The capabilities of Agents and DTs were identified based on text passages describing the implementation of the paradigms in the corresponding reports. This analysis focused on the description of the actual implementation of Agents and DTs, usually found in the methodology section of a report. This means that capabilities which were addressed in the introduction of a report but not in the section describing the actual implementation were not considered.

The set of capabilities exhibited by Agents and DTs is presented in Table 4. Initially, the set of capabilities was iteratively adjusted to encompass all the functions and behaviors of Agents and DTs. The set of capabilities was created irrespective of the paradigm, i.e., no capability is considered as exclusive to one paradigm a priori. The exact definitions of the capabilities as applied in this work are stated in Tables 6, 7, 8 and 9.

The following text passage provides an example of identifying the capabilities exhibited by a MAS. In the report, Agents are utilized as a decision support for human operators in planning the adaptation of production resources. The passage describes how different options for adapting the production resources to fulfill a given product order are presented to a human operator: “*To support the decision maker in the last phase of the adaptation process, the previously found adaptation options (**'flexibility'**) are compared by their KPIs and only a preselection (**'prioritization'**) of adaptation options is given to the user (**'human interaction'**) who is then capable of selecting the most appropriate one depending on the context. The preselection is done via negotiation (**'negotiation'**) between the different debater roles (**'internal system interaction'**) in the Agent system. Suitable approaches for that can be found in the field of multi criteria decision making (**'decision making'**). In most cases, the selected adaptation option by the user requires the adaptation (**'adaptability'**) of the underlying model of the manufacturing machines.*” (Marks et al., 2017) It is worth noting that this text passage contains a notably high number of indicators of capabilities and was chosen for illustrative purposes.

It was assumed that the MAS described above exhibits the capabilities highlighted in the quoted text passage. No distinction was made regarding the level of detail or the number of indicators for each capability. Thus, an application of Agents or DTs either exhibits a given capability (1) or does not exhibit it (0).

Identification of properties

Four properties were evaluated: *autonomy*, *intelligence*, *sociability*, and *fidelity*, as defined in Table 5. The decision to differentiate between *intelligence* and *autonomy* as separate categories was made due to the fact that there can be techni-

Table 4 Capabilities for the comparison

Adaptability	Condition monitoring	Context-awareness
Cooperation	Coordination	Decision-making
Encapsulation	External-system-interaction	Flexibility
Human-interaction	Internal-system-interaction	Learning
Mobility	Negotiation	Prediction
Prioritization	Proactivity	Reactivity and External-system-control
Real-time capability	Robustness and Resilience	Simulation
Synchronisation	Uncertainty-handling	Visualisation

cal systems that exhibit *intelligent* and partially *autonomous* behavior while still being fully controllable by external sources (see also (Müller et al., 2021) for this distinction). However, the fact that a system is controllable from the outside contradicts common definitions of *autonomy*. Another example is an *autonomous* system that is unable to cooperate with its peers. The system might be very *autonomous* in the sense that it can handle unforeseen events etc. but it lacks capabilities in the area of *sociability*. Thus, it makes sense to analyze *sociability* and *autonomy* separately. The different properties are not mutually exclusive and partially overlap with each other. The chosen properties are not exclusively associated with one paradigm, e.g., *intelligence*, which is a key property of Agents, is also increasingly exhibited by DTs (Ashtari Talkhestani et al., 2019).

To facilitate a comparison of the properties of Agents and DTs on an ordinal scale, a property fulfillment score is introduced and determined. The property fulfillment score is calculated based on the previously defined set of capabilities. For each capability, it was determined whether and to what extent it contributes to the fulfillment score of a given property. This determination is inspired by established measures of the degree of *autonomy* in technical systems, which are described in the following paragraphs. Similarly, the relevant capabilities for the properties of *intelligence*, *sociability*, and *fidelity* could be determined and their contribution to the property fulfillment score could be established.

There are several attempts, mainly domain dependent, to determine the *autonomy* of a system:

In 2007, the “ALFUS framework” was published, which defined a framework for *autonomous* capabilities of unmanned aerial systems. Huang et al. (2007) propose measuring the *autonomy* of the system based on the decreasing level of human intervention required. The levels of robot *autonomy*, described by Beer et al. (2014), consider the degree of required human intervention in sensing, planning, and acting primitives. The authors propose nine levels, ranging from manual task execution to full *autonomy* within each of the primitives (Beer et al., 2014).

The SAE standard J3016 describes different levels of *autonomy* for *autonomous* driving systems (Shi et al., 2020). These levels are based on the allocation of responsibility

Table 5 System properties and their descriptions

Property	Description
Fidelity	Jones et al. (2020) define fidelity by the number of parameters exchanged between the physical and the virtual system as well as parameter accuracy and their level of abstraction. A high fidelity virtual model is designed to accurately replicate the physical entity and to realistically correspond to the system (and its behavior) it is created to emulate (Jones et al., 2020; Maier et al., 2017)
Intelligence	KrishnaKumar (2003) defines that intelligent systems emulate aspects of intelligence observed in nature. This includes the ability to process experiences and learn from them, the ability to store knowledge, and the ability to recognize new situations and react to them. Intelligent applications are usually associated with capabilities such as decision making, process control, recognition of patterns and their environment as well as a degree of autonomous behavior in, e.g. maneuvering unknown environments (Kruse et al., 2022)
Autonomy	Autonomous systems can be defined as systems which achieve their objectives within an uncertain environment systematically and without external intervention. (Müller et al., 2021)
Sociability	The property to communicate, interact and coordinate among distributed systems is often called sociability. A paradigm exhibiting sociability is able to transfer data or information using communication protocols and moreover can negotiate with other systems (Han, 2012)

and task execution that are either executed by the driver or the automation system. The tasks include motion control, sensing of the environment and all necessary sub-tasks to fulfill a secure transportation mission. The increasing *autonomy* mainly refers to the decreasing reliance on a driver as a fallback solution in case of unforeseen events and operating in unfamiliar domains. The levels range from 0 (no driving automation) to 5 (full driving automation). The “Industry 4.0 platform” has adapted these levels to industrial production applications, focusing on the degree of required assistance by the user as well (Federal Ministry for Economic Affairs and Energy, 2019). Therein, the “Industry 4.0 platform” pro-

Table 6 Capabilities within autonomy

CD	Capability	Definition
1	External-system interaction	The system is able to perceive (Müller et al., 2021) and exchange information with external systems (applications, services, users, resources, systems,...) (Boussaada et al., 2016)
1	Encapsulation	The system's inner states, functions, goals, and strategies are hidden from the outside and cannot be accessed by external entities (Vogel-Heuser et al., 2020; Wooldridge, 2002)
2	Reactivity and external system control	The system is able to react in a timely and suitable manner to the data and information it perceives (Müller et al., 2021). The reaction can have a physical effect within the system's domain, i.e., it can use physical actors if necessary. This is in contrast to actions that solely change, for example, the inner states of a controller (Bussmann et al., 2004)
3	Decision making	A decision is the choice between different alternatives (incl. the option not to act at all) (Laux et al., 2018). There must be at least two alternatives between the autonomous system can choose. At least two of these alternatives must be decided by the fact that one alternative achieves a certain goal better or worse (Laux et al., 2018). In other words, the alternatives must have different effects (Bussmann et al., 2004) or achieve the same effect with a different degree of efficiency (Müller et al., 2021)
3	Mobility	An autonomous (software) system is able to transmit itself, i.e., its program and states, across a computer network and recommencing execution at a remote site (Wooldridge, 2002)
4	Flexibility	The system is capable of using multiple ways of achieving its goals (Padgham & Winikoff, 2004) as well as deliberating about which goals to pursue (Braubach et al., 2005)
4	Proactivity	The autonomous system chooses its actions according to its goal (VDI, 2010). Such an explicit goal orientation often requires some kind of strategy (Wooldridge & Jennings, 1995) and is more complex than the implicit design objective of, for example, a simple if-then rule-based controller. This also requires the capability to plan and select the most appropriate actions according to a certain goal (Müller et al., 2021). As every system is designed to meet certain design objectives, the autonomous system needs to be able to (pro)actively pursue these goals without being triggered by external events or entities (Müller et al., 2021; Vogel-Heuser et al., 2020)
4	Adaptability	An autonomous system is adaptable if it can change its own structure, shape or behavior (Müller et al., 2021)
4	Robustness and resilience	The autonomous system can maintain its system states and functionality under the influence of internal and external disturbances (Gorodetskii, 2012) and withstands high impact disruptive events (Dehghanpour & Nehrir, 2018)

vides an indication of which capabilities a system requires to achieve a given level of *autonomy*.

The above mentioned frameworks associate the level of *autonomy* that a system exhibits with the degree to which the respective system is capable of executing autonomous behavior. Thus, as a higher level of *autonomy* requires more complex capabilities, a system with a higher level of *autonomy* reflects the possession of more complex capabilities. To avoid the pitfall of associating a high level of autonomy to a system that possesses a number of complex capabilities but lacks other, more basic ones, the methodology of this contribution directly maps the capabilities to so called Complexity

Degree (CD). The CDs are integers from 0 to 5. They rate how complex the implementation of a given capability is and how strongly it contributes to the property fulfillment score of each of the four properties. Thus, each capability is analyzed in the context of each property. It should be noted though that each of the four subsets of the different capabilities and associated CDs does not contain all 26 capabilities, since not every capability is relevant to the respective property.

The fulfillment score of a property is calculated using Eq. (1). In Eq. (1), x is the number of capabilities which contribute to the property, n_i represents the complexity degree of the i -th capability, and $cap_{p,i}$ takes the value 1 if the i -

Table 7 Capabilities within sociability

CD	Capability	Definition
1	External-system interaction	see Table 6
1	Human-interaction	Human interaction refers to systems that offer visual representations of a system's behavior or personalized recommendations to users (Pires et al., 2021)
1	Internal-system interaction	A system which is able to interact with other systems of the same type is capable of the internal-system-interaction (Boussaada et al., 2016)
2	Coordination	The ability to organizing and aligning the actions of multiple entities towards a common objective while minimizing the use of resources. (González-Briones et al., 2018)
3	Cooperation	The ability to collaborate with other systems to achieve goals that may be beyond the capabilities of individuals alone (Wooldridge, 2002)
3	Negotiation	The system is capable of interacting with other systems that may have conflicting interests, in order to negotiate and reach mutually acceptable agreements among all parties involved (Etukudor et al., 2020)

th capability is exhibited by the Agents or DTs in report p and takes the value 0 otherwise. This means that a property fulfillment score of 100% is achieved if all capabilities contributing to the property are exhibited by the Agents or DTs in a given report.

$$\frac{1}{\sum_{i=0}^x n_i} \cdot \sum_{i=0}^x \text{cap}_{p,i} \cdot n_i \quad (1)$$

In the following, each property and the underlying capabilities that contribute to property fulfillment scores of all four properties are described and assigned a CD based on a short literature analysis.

Autonomy and its associated capabilities

The set of capabilities which contribute to the *autonomy* of a system is based on the definitions reviewed by Müller et al. (2021) who argue that the absence of human intervention, self determination, perception of the environment, decision making capabilities, goal orientation and the ability to adapt as well as proactive planning and execution of actions are important aspects of a system's *autonomy*. This is in line with other definitions, for example in Vogel-Heuser et al. (2020) or Meyer et al. (2009). Therefore, the capabilities listed in

Table 8 Capabilities within intelligence

CD	Capability	Definition
1	Condition monitoring	Capability of dynamic representation by an automated, at least one-way data connection to the physical system (Kritzinger et al., 2018)
2	Context awareness	A paradigm is context-aware when it utilizes contextual data to provide the user with significant information or services that are dependent on the user's task (Dey, 2001). In addition, the system should be capable of detecting changes in environmental conditions and effectively utilizing them in accordance with varying requirements, thus contextualizing them appropriately (Rosenberger & Gerhard, 2018)
3	Decision making	see Table 6
3	Sociability CD 1	External-system-, Internal-system-, and Human-Interaction (see Table 7)
4	Prioritization	A system providing prioritization should be able to classify incoming messages and tasks based on their relevance in achieving its own goals (Schoenwald et al., 2004)
4	Uncertainty handling	Uncertainty handling refers to the ability of a paradigm to effectively manage imprecise, contradictory, uninterpretable, and even missing contextual information (Böhm et al., 2021). Even though different uncertainties can be faced by a paradigm in production settings, all kinds of uncertainties are considered under this capability
4	Flexibility	see Table 6
4	Sociability CD 2 & 3	Coordination, Cooperation, Negotiation (see Table 7)
4	Prediction	Capability to forecast future system data output based on historical data or external input (Birk et al., 2022)
5	Learning	The system possesses the capability to learn from known situations/experiments and adapt its behavior accordingly (Boussaada et al., 2016)
5	Autonomy CD 4	Proactivity, Adaptability, and Robustness (see Table 6)

Table 6, sorted by their associated CD, have been extracted from the literature as contributors to *autonomy*.

Sociability and its associated capabilities

Distributed systems should possess the ability to optimize their interactions and communicate with other systems. According to Wooldridge (2002), *sociability* can be classified into different degrees, which determine the complexity

Table 9 Capabilities within fidelity

CD	Capability	Definition
1	Visualization	Capability to graphically represent the system or its behavior (Marcus et al., 2003)
2	Condition monitoring	see Table 8
3	Real-time capability	Capability to react under tight temporal restrictions (Julian & Botti, 2004) which are predefined by the physical system
3	Synchronization	Capability to recognize and react to changed data in the real system in order to uphold its consistency to it (Ashtari Talkhestani et al., 2018)
4	Prediction	see Table 8
5	Simulation	Capability to describe the physical system via mathematical models or a simulation environment in order to enable a reflection of dynamic system behavior (Kritzinger et al., 2018; Sahlab et al., 2022)

of the capabilities described in this work. Table 7 defines the set of capabilities of *sociability*.

Intelligence and its associated capabilities

Meyer et al. (2009) provide an overview of intelligent products, define *intelligence* and present levels for the classification of intelligent products. In this work, these findings are generalized to intelligent production resources. The minimum requirement to be classified as *intelligent* is the ability to handle information sent by external systems, such as sensors. Products that can derive problems from the received information are classified at the second level of *intelligence*. At the third level, resources possess the degree of *intelligence* to manage themselves and make their own decisions. These levels of product *intelligence* serve as a basis for determining the CD of the capabilities associated with *intelligence*. Table 8 defines which capabilities constitute *intelligence*.

Fidelity and its associated capabilities

Both Agents and DTs should possess the ability to accurately analyze, describe, and reflect the physical system(s) they are associated with. This often requires a high *fidelity* of models used to describe the system. Table 9 lists the capabilities of which *fidelity* consists.

Identification of purposes

The utilization of Agents or DTs in the context of production always serves a specific purpose. Therefore, based on the literature analysis presented, several categories of purposes were identified. These categories are the main focus when applying Agents and DTs in the context of industrial automated production systems. The analysis conducted for the first research question (“**Motivation**” section) aims to identify whether certain purpose categories are more commonly associated with one paradigm or the other. This will subsequently address possible correlations between required capabilities and specific purpose categories. The analysis primarily focuses on the life-cycle phases of commissioning and operation of existing applications of Agents and DTs. To identify these purpose categories, an initial list was compiled based on the existing literature, and this list was further extended and updated during the initial screening process.

For the initial definition of purpose categories, the work of Leitão (2009) and Colombo (1998) was consulted. These works introduced an architecture for manufacturing control systems, and the individual elements of this architecture were interpreted as purposes in this work. The major purpose categories in manufacturing control systems include *planning*, *scheduling*, *dispatching* and *control*. Typically, these tasks are executed sequentially in the mentioned order. *Monitoring* is used to track essential information about processes, products, and resources from the shop floor or manufacturing system. This information is then used as a feedback loop to improve *planning*, *scheduling*, *dispatching*, and *control* activities (Leitão, 2009). The results of *monitoring* can also be utilized for diagnosis, error identification and correction, and predictive maintenance which are grouped under the category *diagnosis and fault management*. Other purpose categories frequently associated with Agents and DTs, but not covered by Leitão (2009), include *user assistance*, *virtual commissioning*, and general *process optimization*. Table 10 provides a summary of the individual purpose categories along with a brief description. The purpose of an application involving Agents or DTs was determined by the reviewer after thoroughly reading a report and assigned to one of the nine purpose categories listed in Table 10.

Table 10 System purposes and their descriptions

Goal category	Description
Planning	Planning includes the decision of when and in what quantities a product is to be produced and by when the products are due. In Planning, possible operations to produce the products are derived, required machine capacity and material quantities are evaluated, (Prashar, 2023) and resources are assigned to operations (Colombo, 1998). In this work, the planning of reconfiguration actions are also included in planning
Scheduling	Scheduling deals with the issue of what action a given resource performs at what time. Scheduling is performed well ahead of the actual manufacturing (days or even weeks). The constraints of the resulting schedule are predefined by the planning. (Colombo, 1998)
Dispatching	In dispatching, production resources are coordinated to fulfil a job that has already been released onto the shop floor. Therein it is decided what job a resource will perform next, taking into account the current status of available resources. (Leitão, 2009)
Control	Control is considered to be the process of receiving sensor signals and applying a given control logic in order to send the required actor signals. Control happens in real time. (Colombo, 1998)
Diagnosis and fault management (error recovery/predictive maintenance)	Diagnosis and fault management deal with the evaluation of undesired behavior of resources to identify its source (e Oliveira et al., 2023). Adequate reactions to abnormal behavior are also determined in this category (Colombo, 1998). In this work, we consider predictive maintenance to be included in diagnosis and fault management
User assistance	User assistance includes use cases where a computer program is used to proactively assist a user working on a given problem (Wooldridge, 2002). The problems can fall within one of the other categories presented here, so the focus is on the interaction of the user with a computer program
Monitoring	Monitoring provides other use cases like planning or dispatching with the necessary information. This includes the state of production resources and the progress of operations (Colombo, 1998). The health of a resource or a set of process parameters can be the captured in monitoring
Virtual commissioning	Virtual commissioning aims at the verification of manufacturing systems by connecting a virtual model of a production resource with a real controller or a suitable controller model, thus allowing to simulate and verify both resource and controller behavior. (Lee & Park, 2014)
Process optimization	Process optimization has the goal of improving the design of a new resource or the performance of a given resource. The improvements can lead to increased resource efficiency or production volume (Terwiesch & E Bohn, 2001). The focus of process optimization as applied in this work lies on electrical, mechanical, chemical or biological processes. Therefore, the adjustment of process parameters or the replacement of components can be the result of process optimization

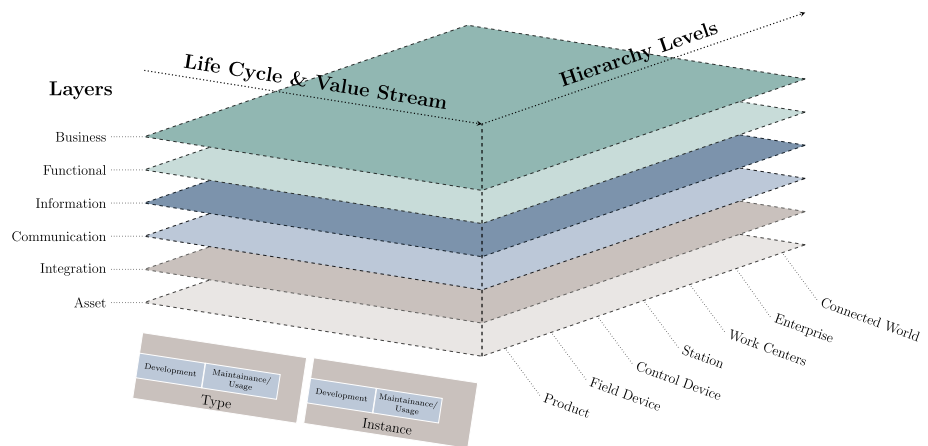
Allocation of applications in the reference architecture model industry 4.0

The use cases described in the reports were further classified based on the individual elements of the axes of RAMI 4.0 (DIN, 2016) (see Fig. 1). This classification was performed to enable both the categorization and aggregated comparison of all use cases for each paradigm. Since both Agents and DTs are widely applied in Industry 4.0 use cases, RAMI 4.0 serves as a suitable framework for classifying these paradigms. RAMI 4.0 allows for the description of an Industry 4.0 component throughout its lifecycle within the context of Industry 4.0 (DIN, 2016). The classification within RAMI 4.0 aims to reveal insights about frequently mentioned layers, hierarchy levels, and life cycle phases in applications of each paradigm. A comprehensive description of the six layers, seven hierarchy levels, and four life cycle phases can be found in (DIN, 2016).

To allocate reports describing the application of Agents and DTs within RAMI 4.0, an examination was conducted to determine whether a specific element of one of RAMI 4.0's three axes was relevant to the application described in the respective report. For example, a single element could be the *functional* layer, at the *work center* level in the usage phase of an *instance*. If a particular element was identified as being of significant importance for the use case, it was marked as the “focus element”. Elements that were addressed within a report but were not considered the focus of the report were still extracted to analyze the application of the paradigms across all elements.

As the focus of this work is the analysis of implemented Agents and DTs, all implementations are considered to be *instances*. Therefore, the elements of the “Lifecycle & Value Stream” axis are always considered as an *instance* in the usage phase. Consequently, the *lifecycle and value stream* axis was not explicitly extracted from the reports and will not be discussed in (“Results” section).

Fig. 1 RAMI 4.0 (DIN, 2016)



Results

This section presents the results of the analysis conducted using the information extracted from the reports included in the systematic literature review. “[Results of the systematic literature review](#)” section provides a brief overview of the results obtained from the literature review. Subsequently, “[Capabilities and properties of agents and digital twins](#)” section describes the capabilities and properties associated with the applications of Agents and DTs within the reports. “[Allocation of paradigms’ applications in RAMI 4.0](#)” section illustrates the placement of Agents’ and DTs’ applications within RAMI 4.0 and offers an interpretation of the findings. “[Purposes of agents and digital twins](#)” section compares system purposes of applications of both paradigms. A comprehensive overview of all the results can be found in Table 12.

Results of the systematic literature review

As shown in Fig. 2, the three-phase approach outlined in “[Method of the systematic literature review](#)” section, led to the exclusion of 981 records in phase 1 and 344 records in phase 2. The full text review in phase 3 resulted in the exclusion of 90 reports (R1: 41, R2: 27, R3: 22). Consequently, a total of 145 reports were included for the identification of system properties, capabilities, and purposes (as explained in “[Methodology for the systematic determination of capabilities, properties, and purposes](#)” section) and for the subsequent systematic comparison. Out of these 145 reports, 58 reports focus on Agents, 86 reports focus on DTs, and one report focuses on both Agents and DT (Xia et al., 2021). The analysis of the latter report was divided into separate analyses of the DT implementation and the Agent implementation, with the results being incorporated into the overall results of the respective paradigm (see also Table 12).

Capabilities and properties of agents and digital twins

The following sections describe Agent’s and DT’s capabilities (“[Agents’ and digital twins’ capabilities](#)” section) and properties (“[Agents’ and digital twins’ properties](#)” section).

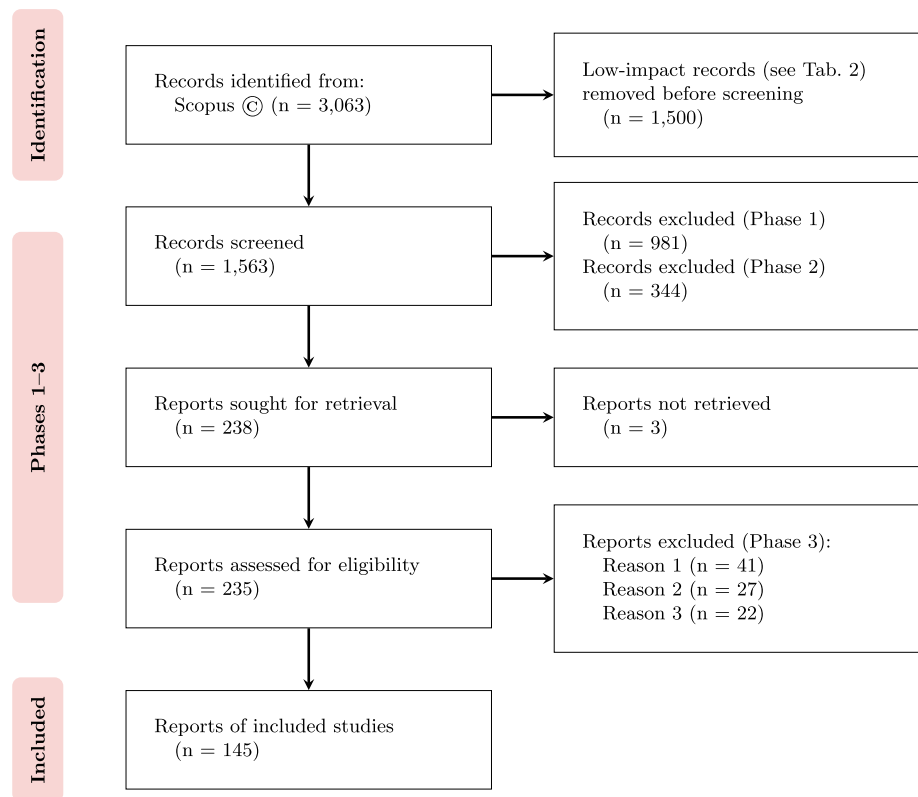
Agents’ and digital twins’ capabilities

Figure 3 shows the rate of occurrence of the investigated set of capabilities in Agents and DTs. The rate of occurrence is calculated by dividing the number of implementations of one paradigm exhibiting a specific capability by the total number of reports focusing on that paradigm. For example, out of 59 reports presenting Agents, 12 implementations exhibit the capability of *learning*, which amounts to 21% as shown in Fig. 3.

The analysis reveals that some capabilities are rarely found in both Agents and DTs. *Proactivity*, *uncertainty handling*, and *mobility* are capabilities that were scarcely addressed in the investigated reports. It should be noted that although Agents and DTs may possess these capabilities, the respective reports did not discuss them, suggesting that they may not be essential for the specific use-cases.

There are also sets of capabilities that are predominantly found in either Agents or DTs. A capability is considered “Agent-specific” or “Digital Twin-specific” if it occurs at a rate of 50% or higher in one paradigm but not in the other. Therefore, the Agent-specific capabilities include *encapsulation*, *internal system interaction*, *context awareness*, *decision making*, *external system control*, *coordination*, *cooperation*, and *flexibility* (in descending order). It is expected that Agents possess these capabilities, as they are commonly applied as MAS, thus requiring capabilities such as *internal system interaction*, *cooperation*, and *encapsulation*. However, the higher occurrence rate of the capability of *external system control* in Agents compared to DTs is surprising, considering that a two-way connection between an asset and a DT is often

Fig. 2 PRISMA 2020 flow diagram according to Page et al. (2021)



described as a requirement in DT definitions (Kritzinger et al., 2018; Tao et al., 2018).

The DT-specific capabilities include *simulation*, *condition monitoring*, *real time capability*, and *visualization*. This observation aligns with common definitions of DTs. However, it is surprising that the capability of *simulation* occurs in less than 70% of DT use cases, considering that *simulation* models are often regarded as a core feature of DTs. Additionally, the capability of *adaptability* occurs less frequently than expected, with a rate of less than 20%. This indicates the challenge of detecting asset changes and adjusting the DT accordingly to maintain accuracy. The same applies to the capability of *synchronization*, which occurs in less than 40% of cases.

This initial analysis reveals substantial differences between the paradigms of Agents and DTs. However, it is also observed that almost no capability is exclusive to one paradigm (with the exception of *mobility*, which was identified in only three applications of Agents, as seen in Fig. 3). This observation is further supported by Fig. 4, which illustrates the occurrence of Agent-specific and DT-specific capabilities in both paradigms.

In the case of Agents, it is observed that the majority possess at least five out of the seven Agent-specific capabilities. Two-thirds of Agents possess at least one DT-specific capability. Similarly, approximately half of the analyzed DTs possess at least one Agent-specific capability. Five DT appli-

cations possess as many as five Agent-specific capabilities. The distribution of DT-specific capabilities among DTs is relatively uniform. Twelve DT applications possess no DT-specific capabilities, while 20 applications possess all four capabilities.

Agents' and digital twins' properties

Figure 5a illustrates the property fulfillment score of DTs for the mentioned properties based on the screened capabilities. It is evident that *fidelity* is the most distinctive property of DTs, as indicated by its median score of 56% and the top quartile at 77% fulfillment.

This aligns with the objective of DT applications, which aim to accurately represent physical assets. On the other hand, *sociability* has a low property fulfillment score, with a median of 9% fulfillment. This is consistent with the definition provided in “*Digital twin paradigm*” section, where social behaviors such as *cooperation* or *communication* are not emphasized. However, it is surprising that *autonomy* also has a low property fulfillment score. Definitions of DTs often emphasize their capability to independently interact with physical assets, but the screening results indicate that this is rarely implemented in practice.

Additionally, *intelligence* has a low fulfillment score, with its median being at 13%. Also in this case, definitions like (Tao et al., 2018; Reiche et al., 2021; Jazdi et al., 2021) claim-

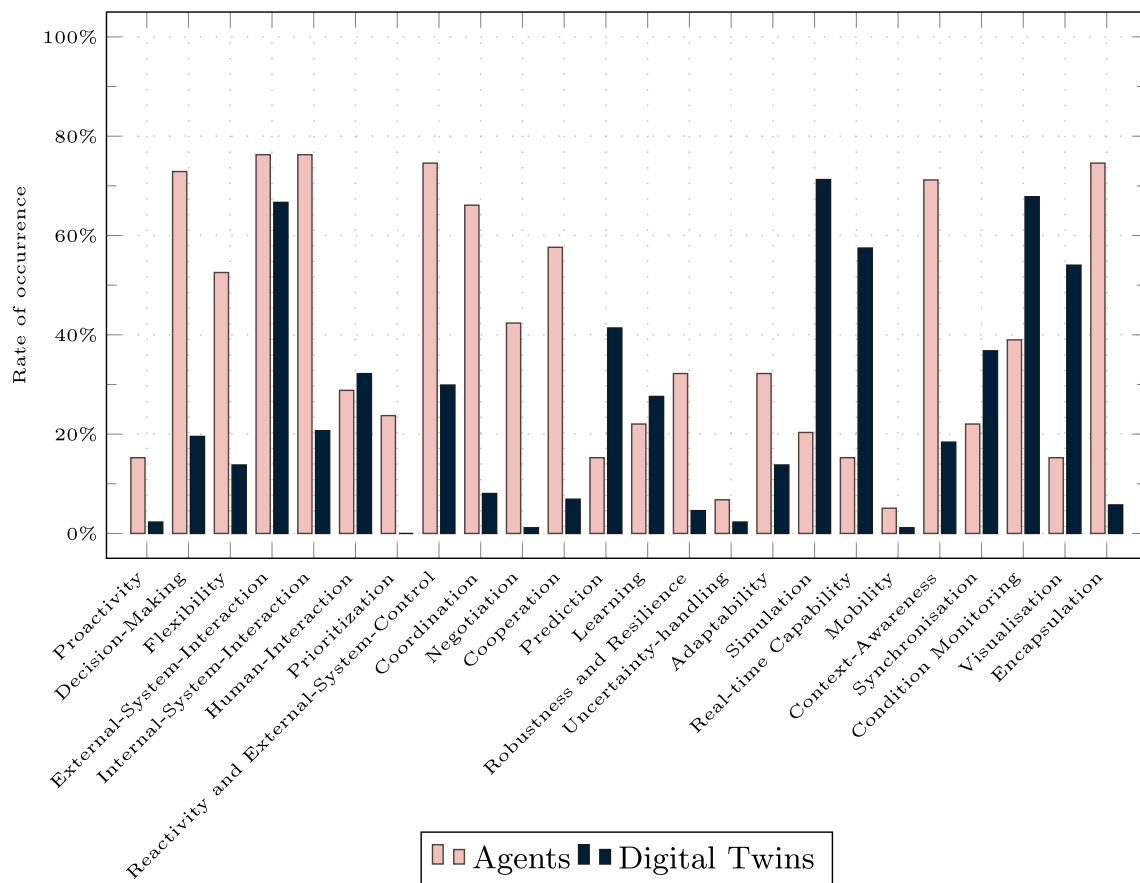


Fig. 3 Rate of occurrence of capabilities in Agents and Digital Twins

ing that *intelligent* components exist within the DT, differ from actual applications.

In contrast, Agents exhibit a different property profile, as shown in Fig. 5b. The majority of the screened applications only demonstrate limited *fidelity*, suggesting that Agents may not be capable of representing physical assets in great detail. However, in terms of *intelligence*, *autonomy*, and *sociability*, Agents exhibit significantly higher scores than DTs.

The data displayed in Table 11 shows for each property and paradigm to what extent the capabilities of each CD have been achieved. For example, a score of 16% for *fidelity* capabilities of CD1 means that on average an Agent-based implementation exhibited 16 % of the CD1 *fidelity* capability subset. As there is only the capability *visualization* for CD1 *fidelity*, this also means that 16 % of the Agent-based implementations exhibited this capability. However, in case of, for example, the two CD3 capabilities for the property *autonomy*, *decision making* and *mobility*, the average score of 39 % for Agents only refers to the average fulfillment and does not necessarily mean that 39 % of the implementations exhibited both capabilities. This is also visible in Fig. 3, showing that *mobility* is less frequent in Agents than *decision making*.

Sociability, which scored the highest with a median of 64 % in reports featuring Agents, is distributed similarly across the CDs (see Table 11). Therefore, most screened reports featuring Agents had the required basic capabilities for interaction, as well as more complex capabilities such as *coordination* and *cooperation*. This finding aligns with the definition provided in “Agent paradigm” section.

The results for *intelligence* in Table 11 show that Agents excel in medium complexity capabilities (CD 2 and 3) such as *decision making* and social aspects, which aligns with common definitions of Agents. However, higher CDs of *intelligence*, involving *learning* and a high degree of *autonomy*, are less frequently achieved. Nevertheless, Agents still score significantly higher in these more complex degrees of *intelligence* compared to DTs.

Autonomy, being one of the core properties of Agents, is also notably strong compared to DTs. However, it is important to note that most approaches exhibit low degrees of *autonomy*, while the higher degrees, consisting of capabilities like *flexibility*, *proactivity*, and *adaptability*, are rarely achieved (see Table 11). This may be due to the divergence between the definitions of Agents and the requirements of production, which was the focus domain of this work. In pro-

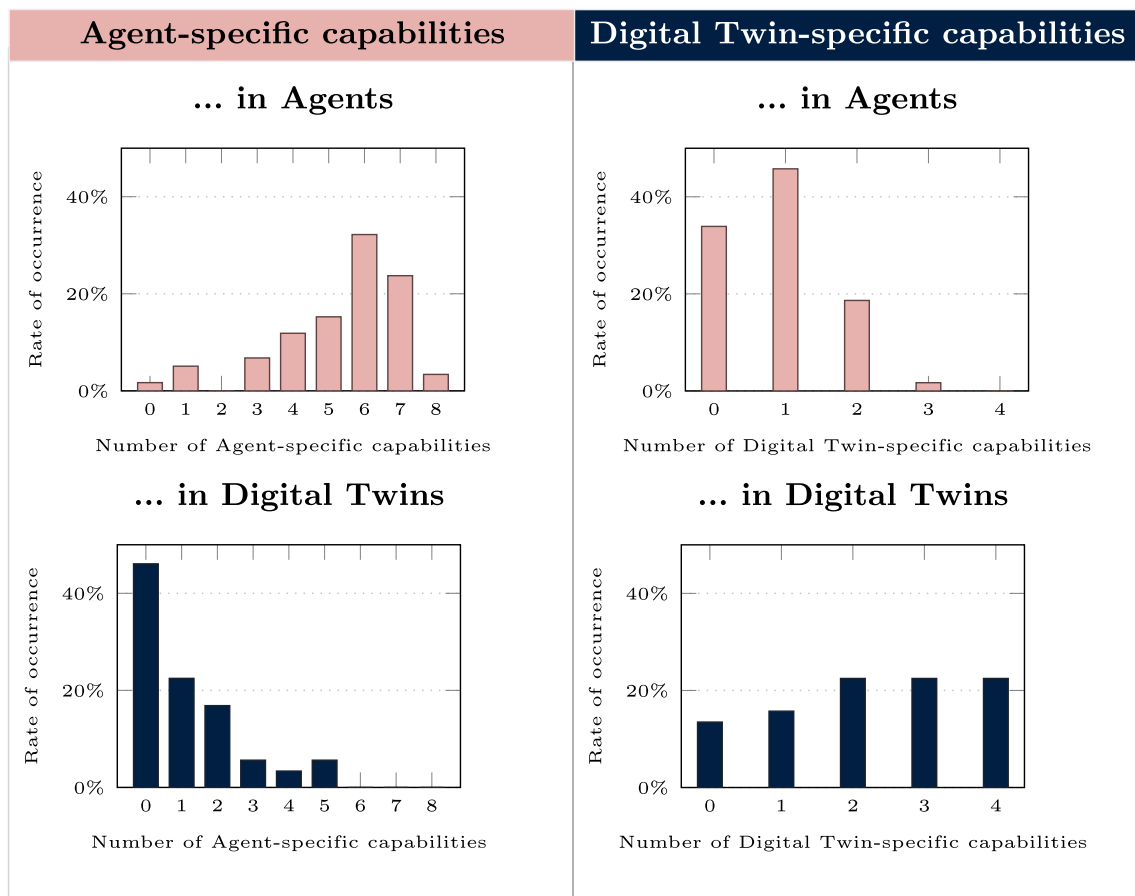
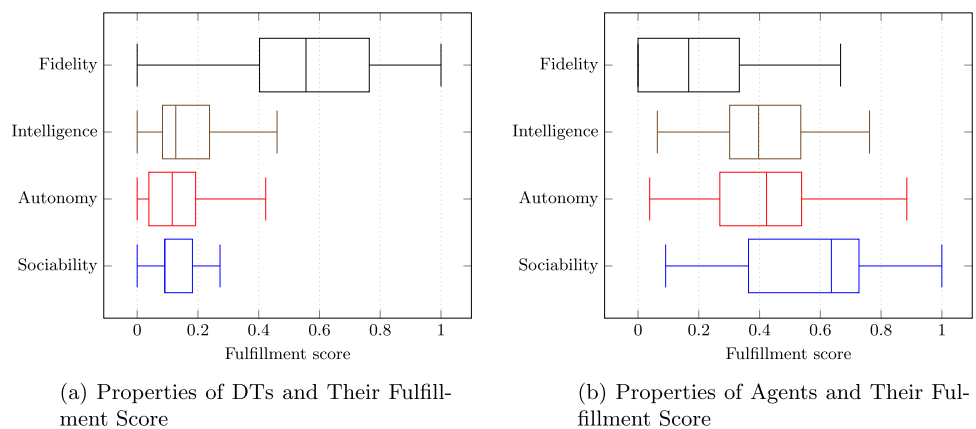


Fig. 4 Comparison of characteristic capabilities of Agents and Digital Twins

Fig. 5 Properties of DTs and Agents and their fulfillment score



duction environments, high degrees of *autonomy* commonly associated with Agent-based concepts are often neither desirable nor necessary. For example, in many applications, Agents are responsible for making decisions based on local information to control the routing of orders and workpieces through the system. Within such constrained environments, it is often enough to incorporate dynamic states of machines and the environment (e.g., *context awareness*) in simple rule based decision processes. This is in contrast to environments

where the Agents encounter situations that are fully unknown and thus no general applicable rules can be provided, corrective actions cannot be taken in time by humans due to latencies, or the environment exposes the Agent to significant threats. In such cases, higher degrees of *autonomy* would be required. Generally, Agents exhibit significantly higher scores of *autonomy* compared to DTs.

Table 11 Average fulfillment score of properties by complexity degree of capabilities

Property	Paradigm		CD 1 (%)	CD 2 (%)	CD 3 (%)	CD 4 (%)	CD 5 (%)
Fidelity	Agents		16	41	20	15	18
	DT	■■■■■	55	68	47	41	71
Intelligence	Agents	■■■■■	41	70	64	38	25
	DT	■.....	68	18	35	10	11
Autonomy	Agents	■■■■■	75	79	39	33	
	DT	■■...■	36	32	11	8	
Sociability	Agents	■■■	61	67	50		
	DT	■...■	40	8	4		

Allocation of paradigms' applications in RAMI 4.0

As described in “Allocation of applications in the reference architecture model industry 4.0” section, part of the analysis is to investigate which elements within RAMI 4.0 are addressed in the analyzed reports. The results of this analysis are presented in the following paragraphs.

Figures 6 and 7 show for each element of RAMI 4.0 at what rate the respective elements occurred as the focus layer of the reports (Figs. 6a, 7a) and at what rate the respective elements occurred within the reports, in general (Figs. 6b, 7b). The value of the rate of occurrence (indicated by the tone of the color in each RAMI 4.0 element) is determined by dividing the number of occurrences in the respective RAMI 4.0 element by the total number of reports focusing on the respective paradigm. The visualization of the results is adapted from Verbeet and Baumgärtel (2020).

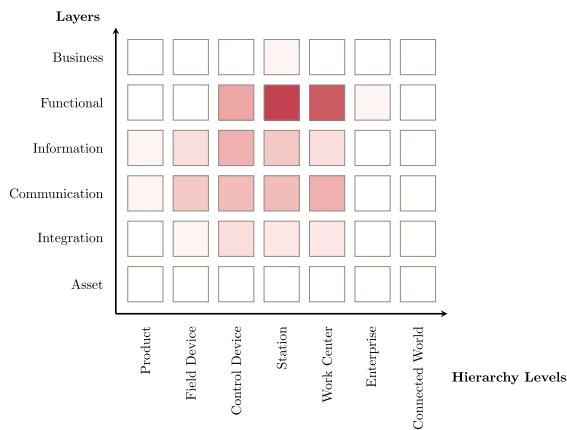
Figure 6 presents the allocation of Agents' applications in RAMI 4.0. Figure 6a displays only the focus elements, showing that the focus of Agents' applications lies primarily on the *integration* to *business* layers as well as the *field device*, *control device*, *station*, and *work center* hierarchy levels. The most frequently selected focus element is *functional – station*. The most frequently selected focus element is *functional – station*. The top three focus elements are found within the *functional* layer, specifically at the *control device* through *work center* hierarchy levels. The *asset* layer is never focused on, nor are the *product* and *enterprise* hierarchy levels, or the *business* layer. This indicates that Agents are primarily applied within industrial production for establishing *communication* between different units of a production system and executing *functions* to facilitate the correct operation of the system. Interestingly, the processing of *information* is rarely the focus of Agents, suggesting that Agents are commonly applied in industrial applications where *information* processing is either a relatively simple task or performed by means other than Agents. As Agents typically do not run on *field devices* or *control devices* (even though such applications exist (Ulewicz et al., 2012)), their frequent allocation to the *station* and *work center* levels are as expected.

Figure 6b depicts the rate of occurrence of RAMI 4.0 elements without differentiation of focus element. It is evident that almost all Agents' applications (94 %) feature the RAMI 4.0 element of *functional – station*, which is the most commonly selected element. The distribution of applications throughout RAMI 4.0 extends to the edges of the axes, indicating that the application of Agents requires consideration of many aspects within RAMI 4.0, essentially up to the *enterprise* level.

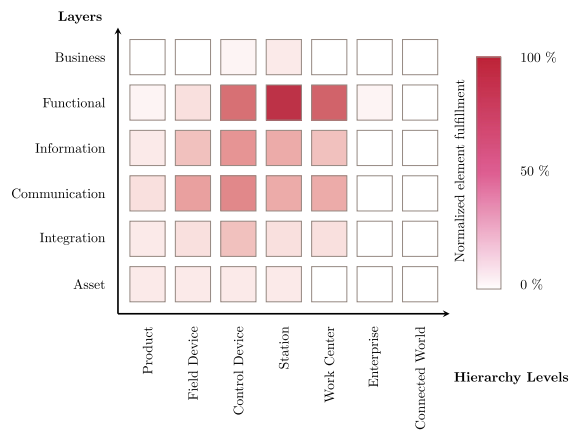
Figure 7 presents the categorization of DTs' applications. As seen in Fig. 7a, DTs show a focus on the *communication*, *information*, and *functional* layers, as well as the *station* and *work center* levels. The combination of the *work center* level and the *information* layer appears to be particularly focused in DTs' applications. However, the surrounding layers are also covered, and there is no layer or level that is not addressed.

Figure 7b displays the categorization of DTs' applications, considering all elements. The results align with the focus element categorization, as the same layers and levels are most commonly found. However, there is a broader spread of observed RAMI 4.0 elements across both layers and levels, covering almost every combination of layer and level. This indicates a wide range of application areas for DTs.

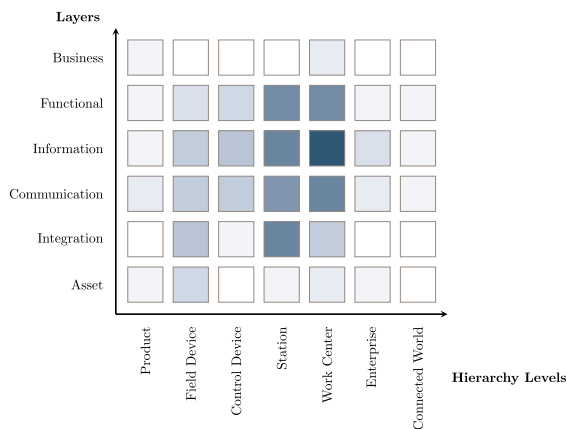
Comparing the placement of Agents' applications (Fig. 6) and DTs' applications (Fig. 7) it is evident that both paradigms mainly focus on the *communication*, *information*, and *functional* layers, with a primary emphasis on the *station* and *work center* hierarchy levels. DTs' applications exhibit a stronger presence towards the edges of the axes compared to Agents' applications. Another interesting observation is that DT applications have a strong focus on the processing of *information*, whereas Agents' applications are primarily focused on executing *functions* within the *planning* and *control* of production processes. This aligns with the definition presented in “Digital twin paradigm” section. However, it is worth noting that for Agents, one might have expected their cooperative and social nature to also benefit use cases in the field of *information* processing and management. Nevertheless, such use cases appear to be less common in practice.



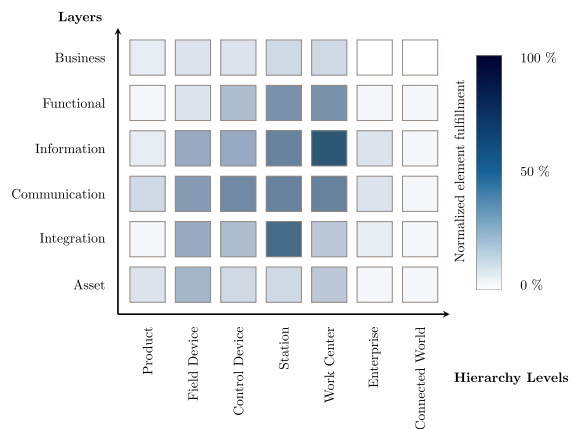
(a) Focus Elements



(b) All Elements

Fig. 6 Placement of Agents' applications in RAMI 4.0

(a) Focus Elements



(b) All Elements

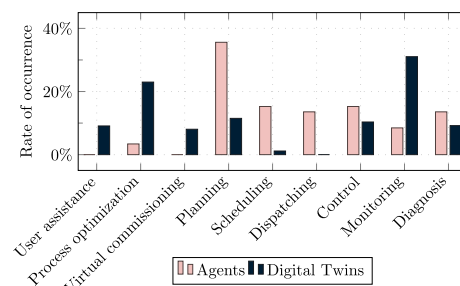
Fig. 7 Placement of Digital Twins' applications in RAMI 4.0

Purposes of agents and digital twins

This section presents the results of the analysis regarding the purposes connected to the applications of Agents and DTs. As described in “[Identification of purposes](#)” section, nine categories of purposes have been identified that Agents and DTs are meant to fulfill: process optimization, user assistance, virtual commissioning, planning, scheduling, dispatching, control, monitoring, and diagnosis and fault management.

Figure 8 shows how often Agents and DTs are applied to serve a purpose each one of the nine categories. Five out of the nine purpose categories are predominantly addressed by one paradigm: *user assistance*, *process optimization*, and *virtual commissioning* are specific to DTs, while *scheduling* and *dispatching* are mostly associated with Agents.

On the other hand, *planning*, *control*, *monitoring*, and *diagnosis and fault management* are areas where both Agents and DTs are applied. This indicates that Agents are primarily

**Fig. 8** Rate of occurrence of purpose categories in Agents and Digital Twins

used during the operational phase of a production system's life cycle, whereas DTs are also utilized in the engineering and commissioning phases, demonstrating the wider range of use cases for DTs observed in “[Allocation of paradigms' applications in RAMI 4.0](#)” section.

During the operational phase, Agents are employed for production *planning*, *scheduling*, and *dispatching*. Production *control* is a category that is evenly distributed between Agents and DTs. *Monitoring* tasks are predominantly performed by DTs, while *diagnosis and fault management* is carried out by both Agents and DTs, with Agents being the more commonly used paradigm. These observations align with the findings from “[Capabilities and properties of agents and digital twins](#)” section, which indicated that there are certain areas where Agents and DTs exhibit strong similarities, while other areas are more specific to each paradigm.

The observation that the DT paradigm is applied to industrial use cases spanning from the engineering phase (*process optimization*) to the operational phase (e.g., *control*) confirms the observations presented in “[Allocation of paradigms’ applications in RAMI 4.0](#)” section, indicating that the DT paradigm encompasses a wider range of elements within RAMI 4.0. Similarly, as discussed in “[Allocation of paradigms’ applications in RAMI 4.0](#)” section, Agents are primarily utilized for executing functions within the *station* and *work center* levels. By referring to Fig. 8, these functions can now be attributed to the domains of production *planning*, *scheduling*, *dispatching*, as well as *diagnosis and fault management*.

Discussion

This work presented a systematic comparison of applications of Software Agents and DTs in industrial automated production systems. The goal of this study was to identify similarities and differences between the two software paradigms, as well as potential synergies between them. The comparison was based on a systematic literature review conducted according to the PRISMA 2020 statement. The focus of the review was on actual implementations of Agents and/or DTs. The two software paradigms were compared in terms of their purposes, properties, and capabilities. Additionally, the study investigated which elements of the RAMI 4.0 architecture are addressed when applying Agents and DTs.

The remainder of this section discusses the methods employed in this study. Subsequently, the results are discussed, and future research opportunities are identified.

Discussion of applied methods

The set of reports analyzed in this study depended largely on the chosen search term and database. *Scopus* was used as experience shows that it yields the highest number of reports. The search term described in “[Methodology](#)” section was formulated to capture reports in the context of automated industrial production systems. Special care was taken to ensure a neutral phrasing of the search term to avoid any

bias towards a specific software paradigm, application, use-case, or RAMI 4.0 element. The balance of 58 reports on Agents to 86 reports on DTs, with one report describing both an Agent’s and a DT’s application (Xia et al., 2021), indicates that this goal was satisfactorily achieved.

It is worth noting that the 3,063 results obtained from *Scopus*, out of which 1500 reports were excluded due to low citation count, represent only a fraction of all research conducted on Agents and DTs in the context of industrial automated production systems. However, it is estimated that increasing the number of reports would have a minimal impact on the quality of the results, as the sample size of 3,063 is already large and the presented results are clear, based on observations that are unlikely to be due to statistical variations. Therefore, future works on comparing Agents and DTs can utilize the same set of reports as this work or follow the same methodology as outlined in “[Method of the systematic literature review](#)” section. However, when applying the methodology in the future, the applied minimum number of citations per year should be carefully adjusted, if reports deemed eligible in this work are to be included even if their number of citations does not increase over time.

Another aspect of the literature review that can be discussed is the number of reviewers per report. Most reports were analyzed by one reviewer, with 15 reports being analyzed by all reviewers to align the review process before the major screening effort. Additionally, 10 % of the reports were analyzed by one additional reviewer after the initial screening to ensure consistency. The results of the comparative screening remained consistent throughout all phases. Minor deviations were observed between individual reviewers, such as whether an implementation of the DT had an individual capability or not. However, these deviations were uncommon, and the results consistently portrayed a clear picture across all reviewers. Having multiple reviewers with different specializations, as was the case in this study, helps prevent bias in a particular direction.

It was decided during the design of this study to only consider reports that presented actual implementations of Agents and DTs for industrial applications. This decision was made to obtain a sense of what kind of capabilities are required and useful in the industrial context and what capabilities are realistically implementable. Purely conceptual reports were omitted because it was deemed that concepts can be proposed irrespective of the need for a concept and its applicability in industrial practice.

To compare Agents and DTs, a set of capabilities, properties, and purposes was created. These sets were developed based on a review of existing literature.

The challenge in establishing a set of capabilities was to encompass all the tasks, features, and functions fulfilled by an Agent or DT. The review process demonstrated that the established list of capabilities adequately met this require-

ment, although it should be acknowledged that the set of capabilities could be adjusted by adding new elements or combining existing ones. An issue encountered in identifying the capabilities of a given application was that reports often use different terminology or certain capabilities were only implicitly observed. In such cases, the interpretation of the reviewer became necessary, and it was in these instances that most divergences between reviewers occurred. A crucial aspect of the chosen approach was to provide clear definitions for each capability, along with corresponding examples, to establish a common understanding among the authors and readers.

The applied methodology for the calculation of the fulfillment score of the different properties enabled a quantitative comparison of the paradigms. It was also utilized to outline the fulfillment of the different CDs within each property. However, the fact that the CD was also utilized to determine the weight of each capability for the property fulfillment score puts a strong emphasis on the more complex capabilities. This emphasis could also be put on the basic or fundamental capabilities, which would lead to different results. The argument for the latter approach would state that, for example, systems are not *autonomous* if they do not possess the basic capability of *encapsulation*. However, the analysis in this contribution followed established metrics like those described in “[Identification of properties](#)” section that emphasise more complex autonomous capabilities.

The set of nine categories of purposes presented in “[Identification of purposes](#)” section fully encompassed the purposes pursued by the analyzed reports. In some cases, multiple categories of purposes were assigned to a single application of Agents or DTs. However, this was only done in a few cases where the authors explicitly stated the application of a paradigm for multiple purposes. On the other hand, some authors did not explicitly state the purpose of the Agents and DTs they implemented. In such cases, the categories of purposes were identified based on general statements in the reports.

In addition to identifying the capabilities of different use cases, a classification of the two paradigms was made in accordance with the different dimensions of RAMI 4.0. The objective was to identify commonalities and differences. Since RAMI 4.0 is a widely recognized reference architecture for Industry 4.0, in which both paradigms are increasingly used, it provided an appropriate basis for the analysis. Furthermore, RAMI 4.0 was considered particularly suitable because it allows multidimensional classification along relevant dimensions. This multidimensional classification facilitated a more comprehensive understanding of these paradigms in relation to Industry 4.0. The use of existing standards to characterize the dimensions promoted the establishment of a uniform and consistent understanding of the paradigms. However, it should be noted that a certain bias

is inherent in the fact that the analysis was limited to the dimensions prescribed by RAMI 4.0. This means that other relevant aspects may have been overlooked. Other architectural models and classifications could have also enabled the classification of the paradigms. Ahleroff et al. (2021) present an architectural model for the DT based on RAMI 4.0. Another architectural model worth mentioning is the “China Intelligent Manufacturing System Architecture”, which is very similar to RAMI 4.0. It consists of dimensions such as “System Hierarchy,” “Lifecycle,” and “Intelligent Functions,” and efforts are being made to align both architectures (Federal Ministry for Economic Affairs and Energy, 2018). It can be assumed that using this alternative reference architecture, which shares strong similarities with RAMI 4.0, would have yielded similar results.

Generally, it would have been desirable to compare the results of this systematic comparison of Agents and DTs with previous, similar works on the topic. However, as stated in “[Related works](#)” section, there exist no other similar comparison-based analyses to which the results could be compared. Despite the lack of comparable analyses, the methodology applied in this work is based on PRISMA and leads to reproducible results.

Discussion of results

The results revealed both differences and similarities between Agents and DTs. Regarding the capabilities of the two paradigms, it was found that *simulation*, *condition monitoring*, *real time capability*, and *visualization* are DT-specific capabilities that imply a high degree of *fidelity*. On the Agent side, the dominant capabilities are *encapsulation*, *internal system interaction*, *context awareness*, *decision making*, *external system control*, *coordination*, *cooperation*, and *flexibility*. This means that Agents tend to achieve medium to high scores in *intelligence*, *autonomy*, and *sociability*.

While both Agents and DTs exhibited paradigm-specific capabilities, all capabilities except for *mobility* occurred in both paradigms. Moreover, more than half of both Agents’ and DTs’ applications possessed capabilities that are characteristic of the other paradigm. This indicates that there is no clear boundary between the two paradigms.

The distribution of capabilities further demonstrated that the typical capability profile of Agents was more uniform than that of DTs. This uniformity was also evident in the analysis of the purposes for which Agents are applied. In the industrial context, Agents are predominantly utilized during the operational phase for *planning*, *scheduling*, *dispatching*, *control*, and *diagnosis and fault management* of production operations. This observation aligns with the allocation of Agents’ applications in the RAMI 4.0 architecture, which shows their primary deployment on the *functional* layer, specifically on computers near the production process

(i.e., at the *station* and *work center* levels). This uniformity can be attributed to the existence of established tools and de facto standards in the field of Agents, such as the *Java Agent Development Framework* and the *Agent Management Specification*. This allows different MAS to exhibit a similar structure which also requires a similar capability set within MAS.

A different pattern emerged in applications of DTs. The purposes served by DTs were more diverse, ranging from the engineering phase (*process optimization*) to real-time *control* of production operations. The allocation of DT applications within RAMI 4.0 also demonstrated greater diversity, with a focus on *information* processing at the *work center* level. Furthermore, the set of capabilities found in DTs was less uniform compared to Agents, as less than half of the analyzed DT applications exhibited over 50 % of the DT-specific capabilities. All of these findings indicated comparatively less standardization and uniformity in DT applications within the industrial sector. This observation is consistent with the conclusions of other authors: *There lacks, [...], a clear, encompassing architecture covering necessary components of a DT to realize various use cases in an intelligent automation system.* (Ashtari Talkhestani et al., 2019). Whether this lack of standardization Similar to the Software Agents paradigm, there is a standard for DT implementations, the AASs. However, this work revealed that implementations of AASs are rare compared to custom DTs implementations.

Summarizing the results and discussion presented earlier, the most significant differences between Agents and DTs align with their definitions provided in “[Agent paradigm](#)” and “[Digital twin paradigm](#)” sections. It was observed that DTs are primarily applied to manage *information* related to production processes and present it to the user, while Agents are mostly employed to plan and manage the operation of production processes. This is evident from the purposes served by the paradigms (Fig. 8) and the capabilities they possess (Fig. 3). However, a clear similarity exists in that both Agents and DTs are software programs that run closely to and interact closely with production resources (Figs. 6, 7). On a more detailed level, it was observed that DTs primarily possess capabilities that enable them to accurately represent production resources, while Agents possess capabilities for cooperative *planning* and operation of production resources (“[Capabilities and properties of agents and digital twins](#)” section). This is also reflected in the properties observed in Agents and DTs (“[Agents’ and digital twins’ properties](#)” section). Furthermore, it is noted that there are almost no capabilities exclusive to either Agents or DTs, and they showed strong resemblance in terms of their ability to interact with other technical systems, interact with human operators, and learn about their environment. Additionally, Agents and DTs are applied for similar purposes,

such as *controlling* production resources, *planning*, *monitoring*, *diagnosing* production operations, and *managing faults*.

Finally, several synergies that represent potential for future research activities can be highlighted. The concept of Industry 4.0 envisions the emergence of *autonomous*, interconnected, and *intelligent* production systems that access, exchange, and process large amounts of data (“[Introduction](#)” section) (Monostori, 2014). Analyzing this vision and the findings of this work, it is evident that aspects related to the *autonomy*, *intelligence*, and interconnection of production systems are primarily addressed by Agents. This includes capabilities encompassing these properties, such as *cooperation*, *decision making*, *reactivity*, and *flexibility* (“[Identification of properties](#)” section). It is worth noting that interconnection, which refers to the degree of interaction between (sub)systems, is part of the *sociability* property as interpreted in this work. On the other hand, other aspects of envisioned future production systems, such as providing real-time data, documenting the evolution of production (sub)systems, and processing *information*, are typically addressed by DTs. DTs commonly employ capabilities such as *simulation*, *real-time capability*, *visualization*, and *condition monitoring*. Based on the analysis presented in this work, it can be concluded that the capabilities and properties of both Agents and DTs should be combined to realize the current vision of future production systems. Agents and their corresponding methods, tools, and standards should be utilized for the *autonomy*-related and cooperative aspects of managing production systems, while DTs provide the necessary informational background and real time connections to their corresponding assets. The analysis of Agents’ and DTs’ properties supports this conclusion, as a combination of Agents and DTs promises high degrees of *intelligence*, *fidelity*, *autonomy*, and *sociability*. Therefore, the existing knowledge on Agents and DTs should be integrated to apply the two paradigms synergistically. Despite the potential of concurrent utilization of Agents and DTs, relatively little research has been conducted on this topic, as demonstrated in “[Motivation](#)” section. However, some works have already leveraged the advantages of both Agents and DTs, similar to the description above (Vogel-Heuser et al., 2021; Göppert et al., 2023). To promote this vision, the connection and collaboration of DTs and Agents need to be standardized.

Summarizing this section, this systematic comparison can serve as a starting point for establishing guidelines on how to apply Agents and Digital Twins in industrial production applications in the future. Based on this comparison and future research, roadmaps can be established for the co-evolution or even the prospective integration of the two paradigms. To foster potential synergies between the two paradigms, identified within this work, their interaction and integration should be standardized. Several purposes served by Agents and Digital Twins (e.g., *planning*, *control*, and

diagnosis and fault management) have been identified, where experiences, pitfalls, and best practices established within one paradigm can potentially be transferred to the other paradigm.

Supplementary information

The dataset related to this article can be found at <https://doi.org/10.5281/zenodo.8120624> (Reinbold et al., 2023).

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Data availability All data generated or analysed during this study are included in this published article (Appendix A). see ‘Supplementary information’ and Appendix A.

Declarations

Conflict of interest The authors have no competing interests to declare that are relevant to the content of this article.

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Appendix A: Comprehensive overview of results

Table 12 shows the results of the analysis of the reports included during the systematic literature review described

in “[Method of the systematic literature review](#)” and “[Results of the systematic literature review](#)” sections. Table 12 first lists all reports which describe an application of Agents and then all reports which describe an application of DTs. If both paradigms occur in one report, it is reported twice. The dataset related to this article can be found at <https://doi.org/10.5281/zenodo.8120624> (Reinbold et al., 2023).

Due to limited space, the following abbreviations are used in Table 12:

Capabilities (Cap.)

Adaptability (Ad), Condition Monitoring (CM), Context-Awareness (CA), Cooperation (Cop), Coordination (Cor), Decision-Making (DM), Encapsulation (Enc), External-System-Interaction (ESI), Flexibility (Fl), Human-Interaction (HI), Internal-System-Interaction (IS), Learning (Lea), Mobility (Mo), Negotiation (Ne), Prediction (Pre), Prioritization (Pri), Proactivity (Pro), Reactivity and External-System-Control (RE), Real-time Capability (RT), Robustness and Resilience (Rob), Simulation (Si), Synchronisation (Sy), Uncertainty-handling (Un), Visualisation (Vis)

Properties

Fidelity (Fid.), Intelligence (In.), Autonomy (Auto.), Sociability (Soc.)

RAMI 4.0 Layers

Focus elements are marked by “(F)” directly following the respective abbreviation. Asset (A), Integration (Int), Communication (C), Information (Inf), Functional (Fun), Business (Bus)

RAMI 4.0 Hierarchy Levels

Similar to the RAMI 4.0 layers, Hierarchy Levels are reported: Product (Pro), Field Device (FD), Control Device (CD), Station (St), Work Centers (WC), Enterprise (Ent), Connected World (CW)

Purposes (Purp.)

User Assistance (UA), Process Optimization (PO), Virtual Commissioning (VC), Planning (Pl), Scheduling (Sc), Dispatching (Di), Control (Co), Monitoring (Mo), Diagnosis (Di)

Table 12 Overview of included reports and corresponding characteristics

Reference	Capabilities	Fidelity (%)	Intelligence (%)	Autonomy (%)	Sociability (%)	Layers	Levels	Purpose
<i>Reports with focus on Agents</i>								
Cagnin et al. (2018)	Ad, CM, CA, Cop, Cor, DM, ESI, FI, HI, IS, Lea, Pri, RE, Si, Vis	44	65	54	73	A, Int(F), C, Inf, Fun	St, WC(F)	Mo
Marks et al. (2017)	Ad, CA, Cop, Cor, DM, Enc, ESI, FI, HI, IS, Lea, Ne, Pri, Si, Un, Vis	33	76	50	100	C, Inf(F), Fun	FD, CD, St(F)	PI
Vrba et al. (2011)	CM, Cor, DM, Enc, ESI, FI, HI, IS, Ne	11	40	42	73	C, Inf(F), Fun, Bus	St(F), WC(F)	Sc
Haben et al. (2021)	CM, CA, Cop, Cor, DM, Enc, ESI, IS, RE, Sy	28	32	27	64	A, Int, C, Inf, Fun(F), Bus(F)	Pro, FD, CD, St(F)	Di
Cimino and Marcelloni (2011)	Ad, CA, Cop, DM, Enc, HI, IS, Ne, Pro, RE	0	46	54	73	Inf, Fun(F)	FD, St, WC(F)	Mo
Dimoulisas et al. (2021)	Cop, Cor, IS, Sy	17	17	8	55	C(F)	St(F)	Di
Khalgui et al. (2011b)	Ad, CA, Cop, Cor, ESI, FI, IS, Pri, RE, Rob	0	54	58	64	Fun(F)	CD(F), WC(F)	PI
Khan et al. (2019)	CM, DM, Enc, IS, Pri, RE, Sy	28	17	23	9	Inf(F), Fun(F)	St(F)	Di
Liu et al. (2014)	CM, CA, Cop, Cor, ESI, FI, IS, RE, Si	39	33	27	64	Fun(F)	St(F)	PI
Merdan et al. (2017)	CA, Cop, Enc, ESI, FI, HI, IS, Ne, Pri, RE	0	43	31	82	Fun(F)	St(F)	Di
Merdan et al. (2011)	Cop, Cor, Enc, ESI, FI, IS, Ne, RE, Rob, Vis	6	43	46	91	C, Inf, Fun(F)	St(F)	Di
Casquero et al. (2019)	DM, Enc, FI, IS, Ne	0	22	31	36	C(F), Inf, Fun(F)	St(F)	Sc
Schutz et al. (2011)	CM, CA, Cor, Enc, ESI, FI, IS, Ne, Pre, RE, RT, Rob, Sy, Un	67	54	46	64	C(F), Inf, Fun(F)	CD(F), WC	PI, Co
Ulewicz et al. (2012)	CA, Cor, DM, Enc, FI, IS, Lea, Pre, RE, RT, Rob	39	48	54	27	C(F), Inf, Fun(F)	CD(F), WC(F)	Co
Vogel-Heuser et al. (2014)	CA, Cor, DM, Enc, ESI, FI, HI, IS, Ne, RE	0	41	42	73	Inf(F), Fun	CD, St(F)	PI
Rodrigues et al. (2018)	Ad, CM, CA, Cor, DM, Enc, FI, IS, Lea, RE	11	43	54	27	Int, C, Inf, Fun(F)	CD(F), St(F)	PI
Rolón and Martínez (2012)	Ad, CM, CA, Cor, DM, Enc, ESI, FI, IS, Ne, Pro, RE, Rob	11	62	88	64	Int, C, Inf, Fun(F)	WC(F)	Sc

Table 12 continued

Reference	Capabilities	Fidelity (%)	Intelligence (%)	Autonomy (%)	Sociability (%)	Layers	Levels	Purpose
Leião et al. (2015)	Ad, CM, CA, Cop, Cor, DM, Enc, ESI, RE, Sy	28	35	42	55	Int, C, Inf, Fun(F)	St(F), WC	PI
Kovalenko et al. (2022)	Ad, CA, Cop, DM, Enc, ESI, FI, HI, Ne, Pri, Pro, RE, Rob	0	67	88	73	Int, C(F), Inf, Fun(F)	St(F), WC(F)	PI
Lepuschitz et al. (2011)	Ad, CA, Cop, Cor, DM, Enc, ESI, FI, IS, Pre, RE, RT, Rob, Si	67	59	73	64	C(F), Inf, Fun(F)	St(F), WC	Co
Bendjelloul et al. (2022)	Ad, DM, Enc, ESI, FI, IS, Ne, RE, Vis	6	35	58	45	A, Int(F), C, Inf(F), Fun	CD(F)	Sc
Deeks (2022)	CA, ESI, Lea, Si	28	16	4	9	Fun(F)	St(F)	Di
Kasprzak et al. (2014)	DM, ESI, FI, Mo, RE, Rob	0	24	65	9	C, Fun(F)	WC(F)	Sc
Kruger and Basson (2018)	Cor, DM, Enc, ESI, FI, IS, RE	0	27	42	36	A, Int(F), C(F), Inf, Fun	WC(F)	Co
Mihoubi et al. (2020)	CA, Cop, DM, Enc, ESI, FI, IS, RE	0	30	42	45	Int, C(F), Inf, Fun(F)	St(F)	Sc
Onori et al. (2012)	Cop, Cor, DM, Enc, ESI, HI, IS, Ne, RE, RT, Rob	17	46	42	100	Int, C(F), Inf, Fun(F)	St(F)	PI
Rehman et al. (2021)	CM, CA, Cor, DM, Enc, ESI, Lea, Ne, Pre, RE, Si	61	41	27	55	A, Int, C, Inf(F), Fun(F)	CD(F), WC(F)	PI
Shin et al. (2019)	CM, CA, Cor, DM, Enc, ESI, FI, Lea, Ne, Pre, Pri, Pro, RE	33	62	58	55	Int, C(F), Inf, Fun(F)	CD(F), WC(F)	PI
Wang et al. (2013)	CM, CA, Cop, Cor, DM, Enc, ESI, FI, HI, IS, Lea, Ne, Pri	11	63	42	100	C(F), Inf(F), Fun	FD(F), CD(F)	PI
Sakurada et al. (2022b)	CM, CA, Cop, Cor, DM, ESI, HI, IS, Pre, Pro, RE, Rob	33	59	54	73	A, Int(F), C(F), Inf(F), Fun, Bus	FD(F), CD(F)	Mo
Adediran et al. (2019)	CA, Cop, Cor, DM, Enc, ESI, IS, Ne	0	37	27	91	Fun(F), Bus	St(F)	Di
Chaplin et al. (2015)	Ad, CM, CA, Cop, DM, Enc, ESI, HI, IS, Pro, RE, Rob, Vis	17	54	73	55	C, Inf(F)	CD(F)	Di
Cicirelli et al. (2018)	CA, DM, Enc, ESI, IS, RE, RT, Si	44	17	27	18	Int(F), C, Fun	CD(F)	Sc

Table 12 continued

Reference	Capabilities	Fidelity (%)	Intelligence (%)	Autonomy (%)	Sociability (%)	Layers	Levels	Purpose
Cruz Salazar et al. (2018)	CA, Cop, Cor, DM, Enc, ESI, IS, RE	0	30	27	64	C, Fun(F)	CD, St(F)	Di
Gräßler and Pöhler (2017)	CA, Cop, Cor, DM, Enc, ESI, IS, Ne	0	37	19	91	Fun(F)	WC(F)	Sc
Hoffmann et al. (2016)	CA, Cor, DM, Enc, ESI, IS, Pro, RE, RT, Rob	17	40	58	36	C(F)	WC(F)	Di
Kang et al. (2022)	CM, CA, Cop, Cor, DM, Enc, ESI, FI, HI, IS, Ne, Pri, RE, RT, Rob	28	63	58	100	Fun(F)	WC(F)	PI
Kattepur et al. (2018)	Ad, CA, Cop, Cor, DM, ESI, FI, IS, Pri, RE, Si, Sy	44	51	54	64	Fun(F)	CD(F)	Co
Kovalenko et al. (2017)	CM, CA, Cop, Cor, DM, Enc, ESI, FI, IS, Pri, Pro, RE, Rob	11	54	73	36	Fun(F)	St(F)	PI
Luo et al. (2015)	CA, DM, Enc, ESI, IS, Pre, Pri, RE, Si, Vis	56	30	27	18	Fun(F)	WC(F)	PI
Marks et al. (2018)	Cop, Cor, DM, Enc, ESI, HI, IS, Ne, RE, Vis	6	38	27	100	Fun(F)	Ent(F)	PI
Natarajan and Srinivasan (2014)	Ad, CM, CA, Cop, Cor, DM, Enc, ESI, HI, IS, Mo, RE, Rob, Sy, Vis	33	52	69	73	Fun(F)	WC(F)	Di
Vallee et al. (2011)	Ad, CA, Cop, Cor, DM, Enc, ESI, IS, Pre, Pri, RE, RT, Rob, Sy, Un	56	65	58	64	Fun(F)	CD(F)	PI, Di
Notomista et al. (2022)	Ad, CA, Cop, Cor, FI, Pri, RE, Rob	0	44	46	45	A, Fun(F)	St, WC(F)	Sc
Frank et al. (2011)	CM, Enc, IS, RE	11	6	12	9	Inf(F)	CD(F)	Di
Gangoiti et al. (2021)	Ad, Cor, DM, Enc, ESI, FI, Ne	0	37	50	55	C(F), Inf, Fun, Bus	CD, St, WC(F)	PI
Jarvis et al. (2018)	Cop, Cor, IS, Mo, RE, Si	28	17	19	55	Int(F), C, Inf, Fun	FD, CD, St(F)	PO
Klein et al. (2018)	CM, CA, Cop, Cor, Enc, IS, Ne	11	29	4	82	C(F)	WC(F)	PO
Khalgui et al. (2011a)	Ad, CA, ESI, Sy	17	16	19	9	A, Int, C(F)	FD(F), CD(F)	Di
Nagorny et al. (2012)	CA, Cop, Cor, DM, Enc, HI, Ne, RE, RT, Rob	17	40	38	82	Int, C, Inf(F)	FD(F), CD(F), St	Co
Rosendahl et al. (2018)	CA, Cop, Enc, ESI, Lea, RE	0	22	15	36	Int, C(F)	FD(F), CD	Co
Švaco et al. (2011)	Ad, CM, CA, Cop, Enc, FI, IS, Ne, RE, Sy	28	37	42	64	C(F)	Pro(F)	PI

Table 12 continued

Reference	Capabilities	Fidelity (%)	Intelligence (%)	Autonomy (%)	Sociability (%)	Layers	Levels	Purpose
Terán et al. (2017)	Cop, DM, Enc, FI, HI, IS, Lea, Ne, Pre	22	48	31	73	C(F), Inf	FD(F)	PI
Wang et al. (2017)	CM, CA, Enc, ESI, FI, RE	11	16	31	9	C, Inf(F)	Pro(F)	Mo
Wein et al. (2021)	CA, Cop, Cor, DM, ESI, IS, RE, Rob	0	38	31	64	Fun(F)	WC(F)	Co, Mo
Bakliwal et al. (2018)	Ad, DM, ESI, Lea, Si, Sy	44	25	38	9	Int(F)	St(F)	PI
Cruz Salazar and Vogel-Heuser (2022)	CM, CA, Cop, Cor, DM, Enc, ESI, FI, HI, IS, Lea, Ne, Pro, RE, Sy, Vis	33	65	50	100	A, Int, C, Inf(F), Fun	Pro, FD, CD, St(F)	Mo
de Giacomo et al. (2023)	CM, Cor, DM, FI, IS, Sy, Un	28	30	35	27	C, Inf(F), Fun	WC(F)	Co, Mo
Xia et al. (2021)	ESI, Lea, Si	0	13	4	9	Inf(F)	WC(F)	Co
<i>Reports with focus on Digital Twins</i>								
Li et al. (2022)	CM, CA, FI, Lea, RE, RT, Si, Vis	61	19	23	0	Int, C, Inf(F)	Pro(F), St	PO
Gichane et al. (2020)	CM, ESI, HI, RE, RT, Si, Vis	61	11	12	18	Int(F)	St(F)	PO
Hoebert et al. (2019)	Ad, CM, DM, ESI, Pre, Si	61	25	31	9	Int(F)	St(F)	Mo
Karanjkar et al. (2018)	CM, Si, Sy	56	2	0	0	A(F), Int(F), C(F), Bus	FD, CD, St, WC(F)	Di
Wenger et al. (2018)	ESI, IS	0	10	4	18	A, Int, C(F), Inf(F), Fun	CD, St(F)	UA
Cattaneo and Macchi (2019)	Ad, ESI, Pre, Si, Sy	67	19	19	9	Int, C, Inf(F), Fun(F)	FD, CD, St(F)	PO
Fernandez et al. (2019)	Vis	6	0	8	0	A, Int, C, Inf, Fun(F)	CD(F), St	Mo
Park et al. (2020)	CM, CA, DM, ESI, HI, Pre, RE, RT, Rob, Si, Sy, Vis	100	33	31	18	A, Int, C, Inf, Fun(F), Bus(F)	Pro(F), FD, WC	PI
Pires et al. (2020)	CM, Pre, RT, Vis	56	8	0	0	Int, C, Inf, Fun(F)	WC(F)	VC, Mo, Di
Jazdi et al. (2021)	ESI, Sy	17	5	4	9	A, Int, C, Inf, Fun(F)	WC(F)	PO
Redelinghuys et al. (2020)	CM, ESI, HI, Pre, RE, RT, Si, Sy, Vis	100	17	12	18	A, Int, C(F), Inf(F), Fun(F)	St(F)	PO
Santos et al. (2022)	DM, ESI, RE, Si, Sy, Vis	50	10	23	9	C, Inf, Fun(F)	WC(F)	VC
Bao et al. (2019)	CM, ESI, RE, RT, Si, Sy	72	6	12	9	C, Inf, Fun(F), Bus	St(F)	
Martins et al. (2020)	CM, Si	39	2	8	0	Int, C(F)	St(F)	UA
Arm et al. (2021)	Enc, ESI, FI, HI, IS, Ne, RE, RT, Si, Vis	50	27	31	55	C(F), Inf, Fun	WC(F)	Di
Cai et al. (2020)	CM, ESI, HI, RT, Si, Vis	61	11	4	18	Int, C(F)	St(F)	Di
O'Sullivan et al. (2020)	CM, ESI, HI, RT	28	11	4	18	Int, C, Inf(F)	St(F)	UA

Table 12 continued

Reference	Capabilities	Fidelity (%)	Intelligence (%)	Autonomy (%)	Sociability (%)	Layers	Levels	Purpose
Zidek et al. (2020)	CM, ESI, RT, Si, Sy, Vis	78	6	4	9	C(F), Bus	Pro(F), St	PO
Protic et al. (2020)	CM, Cop, ESI, FI, HI, RE, RT, Si, Vis	61	24	27	45	C(F), Inf	St(F)	PO
She (2021)	CM, DM, ESI, Lea, Pre, RT, Si, Un, Vis	83	32	23	9	Int, C(F)	WC(F), Ent	Di
Girletti et al. (2020)	CM, CA, DM, ESI, HI, IS, Lea, Pre, RE, RT, Vis	56	38	23	27	A, Int(F), C(F)	St(F), WC	PO
Steringer et al. (2019)	CM, DM, ESI, HI, Pre, Si, Sy	78	22	15	18	Inf(F)	WC(F), Ent(F)	PI
Xie et al. (2022)	Ad, CM, ESI, IS, RE, RT, Si, Vis	61	19	27	18	A, Int, C(F)	CD(F)	PO
Ashtari Talkhestani and Weyrich (2020)	Ad, CM, ESI, IS, RT, Si, Sy, Vis	78	19	19	18	A, Int, C(F)	WC(F), CW	Co
Jeon and Schuesslbauer (2020)	CM, DM, ESI, HI, IS, Pre, RE, RT, Si, Vis	83	27	23	27	C, Inf(F), Fun	WC(F)	VC
Matulis and Harvey (2021)	CM, Cor, DM, Enc, ESI, IS, Lea, RE, Si, Vis	44	30	27	36	C, Inf(F), Fun	St(F)	Mo
Schweizer et al. (2021)	CA, ESI, IS, Vis	6	13	4	18	C(F)	St(F), WC(F)	UA
Rolle et al. (2021)	CM, ESI, IS, RE, RT, Si, Vis	61	11	12	18	C(F), Inf	WC(F)	PI
Mathiesen et al. (2022)	CM, Enc, ESI, RE, RT, Si, Vis	61	6	15	9	C(F), Inf	St(F)	Mo
Zhang et al. (2020)	CM, CA, DM, ESI, IS, Lea, Pre, RT, Si, Vis	83	33	15	18	C, Inf(F)	WC(F)	VC
Azangoo et al. (2021)	CM, ESI, IS, Lea, Pre, Si, Sy, Vis	83	25	4	18	C, Inf(F)	WC(F)	Mo
Martins et al. (2019)	ESI, Si, Vis	33	5	4	9	C(F)	WC(F)	Mo
Ralph et al. (2020)	FI	0	6	15	0	Bus(F)	WC(F)	PO
Caesar et al. (2020)	Ad, FI, Si, Vis	33	14	31	0	Inf(F)	St(F)	PI
Antti et al. (2021)	CM, RT, Si	56	2	0	0	Bus(F)	WC(F)	PI
Garrido and Sáez (2019)	CM, Cor, DM, Si	39	13	12	18	C(F)	WC(F)	PO
Seif et al. (2019)	Cop, Cor	0	13	0	45	A(F)	FD(F), St(F)	Sc
Banunuarachchi et al. (2021)	Ad, CM, ESI, HI, Si, Vis	44	19	19	18	Int(F)	St(F)	PO

Table 12 continued

Reference	Capabilities	Fidelity (%)	Intelligence (%)	Autonomy (%)	Sociability (%)	Layers	Levels	Purpose
Qamsane et al. (2019)	CM, RT, Sy, Vis	50	2	0	0	Int, Inf(F)	St, WC(F)	PO
Tveito and Håkonsen (2022)	ESI, Pre, RT, Si	67	11	4	9	Int(F)	St(F)	Mo
Wenna et al. (2022)	CM, ESI, Pre, RT, Si, Vis	83	13	4	9	Int(F)	St(F)	Mo
Lou et al. (2019)	ESI, HI, Si, Vis	33	10	4	18	Int(F), C	St(F)	Mo
Lugaresi and Matta (2021)	CM, HI, Pre, Si, Sy	78	13	0	9	C, Fun(F)	WC(F)	Mo
Ye et al. (2021)	ESI, RT	17	5	4	9	C(F), Inf(F), Fun, Bus	WC(F), Ent(F)	PO
DallaOra et al. (2022)	CM, Pre, RT	50	8	8	0	Int(F), C(F), Inf(F), Fun(F)	FD(F), WC(F)	PI
Giannetti and Essien (2022)	CM, ESI, Lea, Pre, RE	33	21	4	9	C, Fun(F)	St(F)	Mo
Chirico et al. (2019)	ESI, HI, Si	28	10	4	18	C, Inf, Fun(F)	WC(F)	Co
Schroeder et al. (2016)	CM, ESI, HI, Si, Vis	44	11	4	18	A, Int, C(F), Inf(F), Fun	FD(F), CD(F)	Co
Ward et al. (2021)	CM, ESI, Pre, RE, RT, Si, Sy, Vis	100	13	12	9	A, Int(F), C, Inf, Fun(F)	FD, CD, St(F)	PI
Sierla et al. (2020)	Si, Vis	33	0	0	0	Inf(F), Fun(F)	WC(F)	UA, PO
Schnicke et al. (2020)	Ad, CM, Cor, Fl, HI, RE	11	27	38	27	Int, C, Inf(F), Fun(F), Bus	CD, WC(F)	VC
Bibow et al. (2020)	CM, DM, ESI, HI, Lea, Pre, RT, Sy	67	30	15	18	Int, C, Inf(F), Fun(F)	St(F)	PI
Dario et al. (2019)	HI, Pre, Si, Sy, Vis	72	11	0	9	Fun(F)	FD, St(F), WC	PI
Guerra-Zubiaga et al. (2021)	ESI, Pre, RE, RT, Si, Sy, Vis	89	11	12	9	A, Int(F), C(F), Inf(F), Fun(F)	CD, St(F), WC(F)	PI
Yang et al. (2022)	Pre, RT, Si, Sy	83	6	0	0	Inf(F), Fun(F), Bus	WC(F)	Co
Biesinger et al. (2019)	Pre, RT, Sy	56	6	0	0	A, C(F), Inf(F), Fun(F), Bus	FD, CD(F), St, WC	Co
Nagorny et al. (2012)	CM, Pre, RT, Si, Sy	94	8	0	0	C(F), Inf(F), Fun(F), Bus	FD(F), CD(F), St, WC	Mo
Redeker et al. (2021)	Lea, Pre, RT	39	14	0	0	A, C(F), Inf(F), Fun	WC(F)	PO
Yang et al. (2020)	CM, HI, Lea, Pre, RT, Si, Sy, Vis	100	21	0	9	Int, C, Inf(F), Fun(F), Bus	St(F), WC(F), Ent(F), CW(F)	Co
Kassen et al. (2021)	Cop, Cor, ESI, HI, RT, Si, Vis	50	22	4	64	C, Inf(F), Fun(F), Bus	Pro, CD, St(F), WC(F)	Co
Eckhart and Ekelhart (2018)	HI, Pre, Sy	39	11	0	9	A, Int(F), C(F), Inf(F)	Pro, FD(F), CD(F), St, WC	Mo
He et al. (2019)	CM, ESI, Pre, Si	61	13	4	9	Fun(F)	FD(F), CD(F)	UA

Table 12 continued

Reference	Capabilities	Fidelity (%)	Intelligence (%)	Autonomy (%)	Sociability (%)	Layers	Levels	Purpose
Brosinsky et al. (2020)	Ad, CM, CA, ESI, IS, Lea, RT, Si, Sy	72	30	19	18	C, Inf(F)	CD(F), St(F), WC(F)	Mo
Onaji et al. (2022)	CM, CA, Cop, DM, ESI, Fl, RE, RT, Sy, Vis	50	27	38	36	A, Int, C(F), Inf, Bus	Pro(F), St(F), WC(F), Ent(F), CW(F)	PO
Pérez et al. (2020)	CM, Cop, ESI, RT, Si, Sy, Vis	78	13	4	36	Int(F), C	FD, CD, St(F)	Mo
Ricondo et al. (2021)	CM, ESI, RT, Si	56	6	4	9	Int(F), C, Inf, Bus	FD, WC(F)	PO
Zipper (2021)	CM, ESI, RT, Si, Sy, Vis	78	6	4	9	Inf(F)	CD(F)	UA
Schluse et al. (2018)	CM, CA, Cor, ESI, Fl, HI, IS, Pre, Pro, RE, Si	61	46	42	45	Int(F), C, Inf(F)	FD(F), St(F)	Mo
Leng et al. (2020)	Ad, CM, CA, DM, ESI, Lea, RT, Si, Sy, Vis	78	30	31	9	Int(F), C, Inf, Fun	FD(F), CD, WC(F)	VC
Sjarov et al. (2021)	CM, CA, DM, ESI, Lea, Pro, RT, Si, Vis	61	30	31	9	A(F), Int, C, Inf	WC(F)	Mo
Itigo et al. (2020)	CM	11	2	0	0	A(F)	FD(F)	Mo
Kuhn et al. (2020)	CM, Fl, RT, Vis	33	8	15	0	Int(F), Fun	FD(F)	PO
Kaigom and Rosmann (2021)	CM, CA, DM, Enc, ESI, Fl, Lea, RE, RT, Rob, Si, Vis	61	37	58	9	A(F), Int, C	Ent(F)	VC
Haddod and Dingli (2021)	CM, CA, Fl, IS, Lea, Pre, RT, Rob, Vis	56	38	31	9	Int, C, Inf(F)	St, WC(F), Ent	Mo, Di
Tipary and Erdős (2021)	Ad, CA, Fl, Lea, Rob, Si, Sy, Un	44	40	54	0	Int, C, Inf, Fun(F)	St(F)	Di
Ashtari Talkhestani et al. (2019)	Ad, CM, ESI, HI, IS, Lea, Pre, RE, Si, Sy	78	38	19	27	Int, C, Inf(F), Fun	CD, St(F)	UA
Taha et al. (2022)	CM, CA, IS, Lea, Pre, Si	61	24	0	9	Int(F), C	St(F)	Mo
Ioshchikhes et al. (2022)	DM, HI, Lea, Pre, RT, Si	67	24	12	9	A, C(F)	FD(F)	UA
Juhlin et al. (2022)	CM, DM, Enc, ESI, HI, Lea, Mo, Pre, RT, Si	78	30	31	18	A(F), Int, C	Pro(F)	Mo
Hichri et al. (2022)	CM, ESI, Lea, Pre, Sy	50	21	12	9	A(F)	Pro, FD(F), CD	Di
Yang et al. (2023)	CA, ESI, Lea, RE, RT, Si, Vis	50	16	12	9	Int, C(F)	St(F)	Mo
Czimmermann et al. (2022)	CM, Cop, Cor, HI, IS, RE, RT, Sy, Vis	50	24	8	64	A, Int, C, Inf, Fun(F)	Pro, FD, CD, St(F)	Di
Serik et al. (2022)	Ad, CA, ESI, RE, Si	28	16	19	9	A(F), Int	FD(F), CD	Mo
Siddiqui et al. (2023)	CM, ESI, Lea, Pre, RT	50	21	4	9	Int(F), C, Inf	FD(F)	Mo
Zhang et al. (2023)	CM, ESI, HI, IS, Pre, RT, Si, Sy, Vis	100	22	4	27	A, Int, C, Inf(F)	WC(F)	PO
Caiza and Sanz (2022)	CM, ESI, HI, RE, Si, Vis	44	11	12	18	A, Int(F)	St(F)	Mo
Xia et al. (2021)	ESI, Lea, Si	28	13	4	9	Inf(F)	WC(F)	Co

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